

Three Detectives in a Room: Investigating Character Consistency in Multi-Agent LLM Collaboration

Yifei Chen

March 9, 2026

ACKNOWLEDGEMENT

I thank my instructor Ms. Polina Tsvilodub for encouraging and guiding me on the completion of this thesis.

ABSTRACT

...[TBC] Code and data are available on GitHub. ¹

¹https://github.com/devychen/detective_sim

List of Figures

1	Plot of Predicted Probability and Brier Score for Case 1	32
2	Plot of Predicted Probability and Brier Score for Case 2	32
3	Plot of Predicted Probability and Brier Score for Case 3	32
4	Plot of Sentiment Distance Over Turns For Each Character . . .	43
5	Plot of Sentiment Distance Over Turns For Each Character . . .	43

List of Tables

1	Initial Corpus Statistics and Data Quality Diagnostics (% of Utterances)	23
2	Comparisons of Baseline Performance	24
3	Murderer Identification Accuracy across Three Experimental Conditions	29
4	Regression Results for Predicted Probability of Correct Class . .	31
5	Regression Results for Brier Score	31
6	TF-IDF similarity summary statistics	35
7	Kruskal–Wallis Test Results	35
8	Pairwise Mann–Whitney U Tests Results (Bonferroni Corrected)	35
9	Intra-agent lexical distance summary	36
10	Lexical Consistency Indicators by Character	37
11	Mean Syntactic Depth and Deviation from Literary Baseline by Character	38
12	Pooled linear regression of z-scores (z_{sim}) over dialogue turns . .	39
13	Comparison of Reference- and Simulation-Based Temporal Slopes	39
14	Average sentiment vectors for each character	41
15	Mean Slope of Sentiment Distance for Each Character	41
16	Character-Level Summary of Discourse Consistency Indicators .	42
17	Correlation between linguistic metrics and classifier-based character consistency	44
18	Faithfulness Ranking of Metrics	45
19	Character Contamination	46

Contents

1	Introduction	7
1.1	Application background	7
1.2	Character consistency in linguistic context	7
1.3	Research gap	8
1.4	Research questions	8
1.5	Contribution	9
2	Background and Related Work	9
2.1	Character as a linguistic construct	9
2.2	Character modelling in LLMs	10
2.3	Evaluation of character consistency	11
2.3.1	Human evaluation	11
2.3.2	LLM-as-a-judge	12
2.3.3	Reference-based automatic metrics	12
2.3.4	Psychological and behavioural metrics	12
2.3.5	Task-specific performance evaluation	13
2.4	Linguistic Metrics	13
2.5	LLM-based multi-agent collaboration	14
2.5.1	The evolution of LLM-based multi-agent systems	14
2.5.2	Identity contamination in multi-agent collaboration	15
3	Methods	16
3.1	Prompt engineering	16
3.2	Model selection and inference configuration	17
3.3	Experimental scenarios and case selection	18
3.4	Agent construction: character simulation	19
3.5	Simulation procedure: collaborative reasoning	20
4	Evaluation Scheme	21
4.1	Evaluation Overview	21
4.2	Task performance	22
4.3	Classification baseline	22
4.3.1	Data construction	22
4.3.2	Cleaning and balancing	23
4.3.3	Model Architectures and Training Procedure	23
4.3.4	Baseline performance comparisons	24
4.4	Lexical metrics	24
4.5	Syntactic metrics	25
4.6	Discourse metrics	26
4.7	Validation across metrics	27
4.8	Clustering-based contamination analysis	28
5	Results	28
5.1	Data Overview	28

5.2	Task performance	29
5.2.1	Baseline reasoning capacity (C1)	29
5.2.2	The impact of character prompts (C2)	29
5.2.3	The impact of collaborative dynamics (C3)	30
5.2.4	Case-specific sensitivity	30
5.3	Classifier-based model baseline	30
5.3.1	Holmes	32
5.3.2	Marple	33
5.3.3	Poirot	33
5.3.4	Cross-case summary	34
5.4	Lexical Metrics	34
5.4.1	Vector-based TF-IDF similarity	34
5.4.2	Intra-agent cosine similarity	36
5.4.3	Lexical results summary	36
5.5	Syntactic Metrics	37
5.5.1	Comparison of syntactic depth with literary baselines	37
5.5.2	Temporal trends in structural deviation	38
5.5.3	Comparison of reference- and simulation-based slopes	39
5.5.4	Syntactic results summary	40
5.6	Discourse Metrics	40
5.6.1	comparison of affective profiles with literary baselines	40
5.6.2	Temporal trends in sentiment deviation	41
5.6.3	Visual analysis of sentiment distance	42
5.6.4	discourse results summary	43
5.7	Cross metric validation	43
5.7.1	Lexical level	44
5.7.2	Syntactic level	44
5.7.3	Discourse level	44
5.7.4	Comparative faithfulness ranking	45
5.8	Validation of Character Contamination	45
6	Discussion	46
6.1	Overview of Findings: Addressing Research Questions	46
6.2	a multi-level interpretation of character consistency	47
6.3	Limitation and Future Work	48
7	Conclusion	48
A	Character Prompt	52
B	Rule Prompt	52
C	Case Information	52

1 Introduction

1.1 Application Background

Large language models (LLMs) have achieved remarkable success on a wide range of NLP tasks (text generation, summarisation, instruction following, etc.) [Ji et al., 2025, Guo et al., 2024]. Beyond these general applications, there is growing interest in using LLMs as interactive role-playing agents in simulation environments, entertainment, education, and social settings. For example, LLMs are now being deployed as ‘stand-ins’ for students, patients, voters or citizens in educational tutoring, therapy, social simulation [Park et al., 2022, 2023] and behavioural studies [Aher et al., 2023], and also in entertainments such as games [Xu et al., 2024b]. In such applications, each agent is typically given a persona or fictional character profile that it should be consistently embody. The profile contains role-playing background information such as career, backstory, personality traits, etc. This ability of LLM agents to generate realistic multi-agent dialogues, simulate psychological experiments or drive richly interactive games, opens up novel possibilities at scale and with reproducibility that would be impossible with human actors. For instance, recent work has shown LLMs being used to generate realistic populations in multi-agent social simulations, and to engage in cooperative [Guo et al., 2026] or adversarial [Baltaji et al., 2024] group tasks.

However, a major challenge in these setting is character consistency: an LLM agent must contain its assigned persona throughout a dialogue or collaborative task. In practice, off-the-shelf LLMs often drift from their personas, contradict earlier statements or display inconsistent traits, especially over longer interactions [Choi et al., 2024, Bhandari et al., 2025, Abdulhai et al., 2025]. As Ji et al. [2025] note, current LLMs “lack [...] fine-grained role awareness,” which limits their ability to provide truly personalised, role-consistent interactions. In collaborative multi-agent scenarios, this issue is amplified: agents may influence one another and inadvertently dilute distinct personality features. Maintaining credible character behaviour is thus essential for user satisfaction and system validity in these applications.

1.2 Character Consistency In Linguistic Context

This study approach character consistency from a linguistic perspective. in sociolinguistics and stylometry, a speaker’s or a character’s identity is not viewed as a fixed label but as continuously construction through language use [Przystalski et al., 2025]. In other words, how a character speaks conveys their persona, including choice of words, syntax, discourse style. For example, stylometric studies of television dialogue show that fictional characters develop distinct idiolects: Sheldon Cooper in the show *The Big Bang Theory* habitually uses formal, scientific vocabulary and passive constructions, whereas Penny’s speech is more colloquial and conversational [Van Zyl and Botha, 2016]. Language therefore serves as a window into attitude, beliefs, and personality.

Following this view, this study posits that the character consistency of a LLM-based role-playing agent could be manifested in the linguistic profile of its outputs. If an agent truly embodies a persona, its generated responses should consistently reflect that persona’s linguistic style. In this study, character consistency is therefore treated as fundamentally a linguistic style problem: this study expects to detect (in)consistency by analysing lexical, syntactic and discourse-level patterns in the generated dialogue text. This aligns with sociolinguistic theory that identity is performed and negotiated through language [Bucholtz and Hall, 2005]. By focusing on linguistic features, consistency could be quantified in a principled way.

1.3 Research Gap

Despite the importance of character consistency, existing work on LLM-based agents has largely focused on model design, training paradigm, or learning methods, rather than systematic evaluation. And even evaluation rarely was designed from linguistic perspective. Much prior research has tried to imbue LLMs with persona via instruction fine-tuning such as the character-specific LLMs in CharacterGLM [Zhou et al., 2023] or the Ditto [Li et al., 2021, Lu et al., 2024], chain-of-thought [Chae et al., 2023] or contrastive Reinforcement Learning fine-tuning [Shea and Yu, 2023]. Moreover, they often use coarse or costly evaluation methods: performance is measured by high-level metrics of role-fidelity such as Big Five, Myers-Biggs Type Indicator (MBTI), or human judgements, rather than by analysing the language itself.

1.4 Research Questions

This thesis investigated character consistency in LLM-based agents through the following questions:

1. **RQ1:** To what extent do LLMs maintain character consistency during collaborative reasoning tasks?

Hypothesis: LLMs will partially preserve assigned personas but will exhibit character inconsistency over longer multi-turn dialogues.

2. **RQ2:** How can multi-level linguistic metrics capture character consistency?

Hypothesis: Combining lexical, syntactic, and discourse features will better reflect persona fidelity than any single level alone.

3. **RQ3:** Which metrics are most reliable capture it?

Hypothesis: Metrics that correlate strongly with a dedicated persona classifier’s confidence scores (e.g. classification probability that an utterance matches the persona) will be more predictive of true consistency

These questions guide a comprehensive evaluation of character consistency, from empirical measurement to alignment with predictive models.

1.5 Contribution

In addressing the questions above, this thesis makes the following contributions:

1. **A linguistically grounded multi-level evaluation framework:** This study proposes a systematic evaluation framework that combines lexical, syntactic and discourse-level metrics to assess character consistency in multi-agent dialogue.
2. **Classifier Validation and Calibration:** This study introduces a methodology to validate character consistency metrics by correlating them with a character classification model’s outputs. By treating consistency scores as pseudo-probabilities, this study examines their calibration against classifier confidence, identifying which metrics are most predictive and reliable.

Together, these contributions aim to fill the identified gap: providing a linguistically grounded multi-dimensional evaluation of character consistency in LLM-based agents, especially under collaborative conditions. All components are designed to be extensible for future research in role-playing driven conversational AI.

2 Background And Related Work

2.1 Character As A Linguistic Construct

From the perspective of discourse and sociolinguistic theory, an individual’s identity (in this scenario, a character’s persona) is continuously enacted through language. In sociolinguistics, language is understood to construct identity: speakers adopt styles and linguistic cues that associate them with social groups stereotypes or affective states. Similarly, in narrative, Characterisation is manifested by the way character speak. For example, Culpeper [2001] argued that "an individual’s feelings, behaviours, and cognitive states are represented by and through language", thus language use serves as a primary source of information about a character. In other words, characters are written into existence by their linguistic choices.

This view is borne out by stylometric analyses. Stylometry is the quantitative study of linguistic style, originally developed for authorship attribution. Such methods have been applied to fictional dialogues to distinguish characters. For instance, a stylometric analysis of The Big Bang Theory dialogues [Van Zyl and Botha, 2016] showed the character Sheldon Cooper consistently differs from other characters in vocabulary and grammar: he uses more formal, scientific terms and favours passive voice, giving his language an “expository” quality distinct from his friends’ colloquial style. These distinct styles emerged despite being written by multiple script authors. Similarly, stylometric analysis of Jane Austen novels by Burrows [1987] and others have demonstrated that characters exhibit quantifiable differences in function word usage and lexical

patterns. These examples illustrate that character identity leaves a detectable imprint on language.

Modern stylometry research whether focusing on fictional characters or not reinforces this idea. For authorship attribution or model identification, features such as character n-gram, word frequencies, syntactic structure and sentiments serve as "linguistic fingerprints" of an author or text generator. Studies show that a combination of lexical and grammatical features can distinguish texts written by different LLMs or humans with high accuracy [Przystalski et al., 2025]. They note that LLM-generated text tends to exhibit greater grammatical standardisation and systematic usage patterns compared to human text. These stylistic markers consistently reflect the underlying features of the speaker or model, although maybe subtle.

In sum, the background literature suggests that language style is a valid proxy for character consistency. If an LLM is playing a consistent character, its utterances should collectively form a coherent style profile. Conversely, a sudden in language choices might signal a break in character. This perspective motivates current approach: this study will use linguistic style metrics as indirect measures of character consistency.

In this thesis, the term persona and character is distinguished. This study uses persona to refer to the explicitly assigned profile or identity specification provided to the model through prompting. In contrast, character refers to the identity as realised through linguistic behaviour in dialogue. While persona concerns the intended role, character concerns its performed and observable manifestation. Character consistency therefore denotes the stability of this linguistic realisation across turns.

2.2 Character Modelling In LLMs

In computational linguistics, character modelling typically means conditioning a language model on a character description or background. The simplest method is prompt-based conditioning, where a textual character profile (e.g. "You are Sherlock Holmes, a consulting detective who ...") is prepended to the conversation prompt. In this way, even a general-purpose LLM can be guided to role-play that character. Recent studies confirm that prompting with explicit role profile can effectively steer LLMs. For example, Chen et al. [2023a], Tu et al. [2024] show that chat models given a well-crafted character prompt can emulate that persona in dialogue. This in-context approach leverages the LLM's existing knowledge without fine-tuning, and can produce consistent character-specific responses with minimal examples.

More advanced methods incorporate character information into model architecture or training. Early works without LLMs (with language models for example) fused memory networks or personal databases into dialogue models [Zhang et al., 2018]. With LLMs, researchers have built character-specific LLMs by further training on persona data. For example, CharacterGLM [Zhou et al., 2023] and Ditto [Li et al., 2021, Lu et al., 2024] are instruction-tuned models with enhanced persona consistency for specific characters. Reinforcement

learning techniques have also been applied. For example, Shea and Yu [2023] use RLHF to better align models with persona traits, and Chen et al. [2025] generate preference data with GPT-4 followed by Direct Preference Optimisation (DPO) to refine LLM persona behaviour. These engineering solutions can improve consistency, but they often require expensive data collection or fine-tuning.

Despite these advances, a common observation is that character inconsistency remains a problem. For example, Choi et al. [2024] show that even when LLMs are primed with a persona, they may gradually shift away from it during a long conversation. In their study of nine LLMs, larger models tended to experience greater identity , and simply specifying a persona did not fully prevent its degradation. Similarly, [Frisch and Giulianelli, 2024] found that persona prompts influence language style only up to a point and over multiple turns the models’ output can change qualitatively. This suggests that while persona conditioning methods exist, they do not guarantee indefinite consistency.

In sum, character and persona modelling in LLMs spans a range of techniques from prompt engineering to specialised training. But issue of inconsistency and erosion persist. Many prior works focus on how to inject persona and prevent inconsistency, whereas this study focus on assessing how well the character is maintained in the generated text.

2.3 Evaluation Of Character Consistency

2.3.1 Human Evaluation

Previous work have used a number of approaches. Human evaluation is the most traditional approach and is often regarded as the most reliable form of assessment. Researchers typically recruit human annotators, who may include domain experts or fans of the character being simulated. These evaluators assess the agent’s performance across multiple dimensions. This can provide nuanced feedback, but is costly and subjective.

Common criteria include consistency, which measures whether the agent’s responses align with the predefined character profile, human-likeness, and engagement, which reflects how interesting or immersive the interaction appears to users. For example, Yang et al. [2024], Zhou et al. [2023] report human evaluations of their character customised model CharacterGLM and SimsChat agent for character consistency, and Tu et al. [2024] use human annotators on their CharacterEval benchmark. Zhu et al. [2024] distributed a survey to human participants to capture their gameplay experience and satisfaction levels with agents.

Two common formats are used. In pointwise evaluation, annotators rate the output of a single model according to predefined criteria [Zhou et al., 2023]. In pairwise comparison, annotators compare responses produced by two models and determine which performs better. For example, in role-playing detective games [Zhu et al., 2024], human participants may interact directly with an agent and then evaluate its performance using criteria such as narrative progression

or character immersion.

2.3.2 LLM-as-a-judge

Because human evaluation is costly and difficult to reproduce, recent studies increasingly employ large language models as automated evaluators. In this approach, a stronger model such as GPT-4 is used to assess the responses produced by other agents [Chen et al., 2023b, 2024b, Shao et al., 2023].

Researchers may design structured interview procedures in which the evaluation model asks the agent a series of questions. The responses are then assessed along predefined dimensions, such as memory consistency, value alignment, personality traits, hallucination tendency, and behavioural stability [Shao et al., 2023, Park et al., 2023]. The evaluation model can also rank responses generated by multiple agents according to specified criteria, such as their compatibility with the scenario, character attributes, or interpersonal relationships [Chen et al., 2023b, 2024b].

Borrowing from psychology, some studies apply structured personality tests. Jiang et al. [2024] and Sorokovikova et al. [2024] feed LLM outputs into the Big Five Inventory to score trait alignment. Pan and Zeng [2023] use the Myers-Biggs Type Indicator (MBTI) to categorise LLM "personalities" and observe that each model tends towards a stable MBTI type. Wang et al. [2024c] perform interview-style tests for personality fidelity. These methods can quantify high-level personality consistency, but they rely on the assumption that LLM output can be meaningfully mapped to human trait scales which is still an open question. As for fine-grained error analysis, LLM-based evaluators can further identify specific types of failure. These include out-of-character behaviour (OOC), internal contradictions, and repetitive responses [Zhou et al., 2023].

2.3.3 Reference-based Automatic Metrics

Another line of evaluation relies on existing textual resources or reference answers. One popular dimension is the **text overlaps**. Metrics such as BLEU and ROUGE-L are commonly used to measure lexical overlap between the generated responses and reference texts. In role-playing settings, these references may include original scripts, character dialogue, or curated conversation datasets [Chen et al., 2023b, Wang et al., 2024a,b].

Another dimension is the **semantic similarity** implemented by embedding-based approaches. By computing cosine similarity between generated responses and character descriptions, researchers can evaluate whether the agent’s output remains consistent with the background knowledge or identity profile of the character [Wang et al., 2024b].

2.3.4 Psychological And Behavioural Metrics

In addition to above directions, recent work has begun to evaluate LLM agents from a psychological perspective. One is **psychological scale testing**. Frameworks such as Wang et al. [2024c] assess personality fidelity by asking agents

questions derived from established psychological inventories. Examples include the Big Five Inventory (BFI) or other personality assessment tools [Pan and Zeng, 2023]. The resulting scores are used to determine whether the agent’s responses align with the expected personality traits of the target character.

Another aspect is the **behaviour and motivation prediction**. In narrative or scenario-based environments, agents may also be evaluated based on their ability to predict or simulate the actions of a character. For instance, in detective scenarios, researchers examine whether the agent can correctly anticipate a character’s next move or infer hidden motivations [Shao et al., 2023, Chen et al., 2024a, Wang et al., 2024c, Xu et al., 2024a].

2.3.5 Task-specific Performance Evaluation

When agents operate within collaborative or problem-solving environments, task performance becomes another important evaluation dimension. One metric is the **task accuracy**. In the scenarios similar to this study, that is, investigative games or detective simulations, agents may be evaluated according to their ability to collect relevant clues, identify the culprit, and perform logical reasoning [Wu et al., 2024].

Another aspect is the **team efficiency**. In multi-agent systems, evaluation may also consider group-level outcomes, such as task completion time, communication efficiency, and leadership dynamics within the team [Guo et al., 2026].

2.4 Linguistic Metrics

The methods described above provide useful perspectives on role-playing agents, yet they remain largely fragmented. A unified and multi-level evaluation framework is rarely applied. Even when multiple metrics are used, most operate at the level of individual utterances or predefined personality traits, while paying limited attention to the linguistic form through which character identity is expressed. As Tseng et al. [2024] note in a recent survey, despite the growing number of evaluation approaches, “there still lacks a unifying understanding of how to quantify personality in LLMs”. In addition, existing evaluations seldom examine which metrics most reliably capture character consistency across extended interactions.

To address this limitation, the present study develops a set of metrics grounded in the linguistic hierarchy and validates them through classification performance and confidence calibration. The approach proposed in this thesis differs from existing evaluation frameworks in an important way. Rather than focusing primarily on high-level behavioural outcomes or abstract personality traits, it examines how character consistency is realised through linguistic structure during dialogue. Specifically, the analysis operates across three layers of language: lexical, syntactic, and discourse features.

Most existing studies evaluate role-playing agents through macro-level indicators, such as personality traits, value alignment, or task success. While these approaches provide useful insights, they often overlook how character identity is

realised through language during ongoing interaction [Shao et al., 2023, Wang et al., 2024c, Wu et al., 2024]. The method proposed in this thesis instead investigates the micro-level linguistic structure of agent dialogue, allowing subtle shifts in speaking style to be detected over the course of extended collaborative interactions.

At the lexical level, the analysis can reveal changes in the use of character-specific vocabulary or recurring expressions. Such patterns may indicate whether the agent continues to employ a distinctive character voice [Zhou et al., 2023, Wang et al., 2024a].

At the syntactic level, structural features of sentences can signal shifts in communicative style. For instance, a character originally designed to speak in analytical or elaborate sentences may gradually revert to shorter and more generic responses typical of default assistant behaviour [Park et al., 2023, Wang et al., 2024a].

At the discourse level, the analysis examines how the agent maintains conversational coherence and fulfils its narrative role within collaborative dialogue. This includes how the agent manages topic development, responds to other participants, and sustains the interactional logic of the scenario [Ji et al., 2025].

Although recent work has begun to discuss phenomena such as identity drift or persona drift in interactive agents, these studies often rely on isolated psychological measurements or discrete evaluation snapshots. They rarely track linguistic behaviour continuously throughout the dialogue itself. By contrast, the approach adopted in this thesis provides an intrinsic, corpus-based evaluation framework. Through systematic measurement of lexical, syntactic, and discourse-level features, it offers a new quantitative perspective on how character consistency evolves during long-term interaction [Choi et al., 2024].

2.5 LLM-based Multi-agent Collaboration

2.5.1 The Evolution Of LLM-based Multi-agent Systems

The transition from single-agent systems to Multi-Agent Systems (MAS) marks a pivotal shift in LLM applications, moving from individual task execution to the emulation of collective intelligence [Guo et al., 2024, Xi et al., 2025]. By assigning specialized profiles and enabling inter-agent communication, MAS can effectively mirror human group dynamics in complex problem-solving scenarios, such as scientific debate, software development, and detective reasoning [Guo et al., 2024].

This study chooses the detective case-solving scenario because it represents a pinnacle of complex social reasoning, requiring agents to not only process factual evidence but also to perform Theory of Mind (ToM) inferences regarding the motives and beliefs of other characters [Zhu et al., 2024]. Recent frameworks like WellPlay [Zhu et al., 2024] and Jubensha [Wu et al., 2024] have established benchmarks for measuring the task accuracy and reasoning efficiency of agents in such settings.

However, a significant gap remains in the literature. While current research

heavily prioritizes optimization of cooperation strategies and final task outcomes (e.g., win rates), it often overlooks the linguistic integrity of the personas under the pressure of collaboration [Chen et al., 2024b, Wu et al., 2024, Zhu et al., 2024]. The very nature of multi-round interaction introduces risks of Identity Drift and Persona Inconstancy, where agents may succumb to peer pressure or inadvertently adopt the linguistic patterns of their counterparts [Choi et al., 2024, Baltaji et al., 2024]. Unlike existing works that rely on macro-level performance metrics, this thesis shifts the focus to the micro-linguistic level. By analysing dialogues through lexical, syntactic, and discourse metrics, this study aims to quantify the subtle "linguistic imprint" left by collaboration and detect whether the intense interaction required for problem-solving inevitably leads to character inconsistency.

2.5.2 Identity Contamination In Multi-agent Collaboration

Finally, the special dynamics of multi-agent systems is also worth elaborated. When multiple LLM agents interacts, linguistic phenomena such as alignment and convergence can occur. Recent surveys on multi-agent systems (MAS) highlight that while specialized profiles and inter-agent communication facilitate collective intelligence [Guo et al., 2024], they also introduce complex social dynamics [Zhu et al., 2024]. This includes the risk of information redundancy and "over-reporting," where agents may prioritize group alignment over individual task precision [Guo et al., 2026]. Just as humans in conversation tend to adapt to each other’s style (accommodation), groups of AI agents may inadvertently begin to share vocabulary or syntax. Recent work by Parfenova et al. [2025] demonstrates this: in simulated collaborative annotation tasks, LLM groups were found to lexically and semantically converge over multi-round discussions.

This linguistic convergence poses a severe risk of identity drift and identity contamination. Choi et al. [2024] define identity drift as the phenomenon where interaction patterns or styles change over time, noting that larger models may paradoxically struggle more with identity stability due to their tendency to generate fictitious self-details. Furthermore, Baltaji et al. [2024] identify "Persona Inconstancy" as a critical failure mode in cross-national collaborations, categorising it into conformity, confabulation, and the most disruptive form—impersonation, where an agent abandons its assigned identity to adopt another mentioned in the chat context. Such lapses often stem from the lack of clear separation between role descriptions and the growing linguistic context of the dialogue.

Moreover, the singular persona convergence observed in models fine-tuned via RLHF often leads agents to default to a generic, polite assistant style, eroding the specific values and traits required for complex role-playing [Park et al., 2023, Moon et al., 2024]. Such contamination reduces the diversity and specificity of personas, which undermines the goal of simulating heterogeneous characters. This study argues that traditional macro-level evaluations (like task accuracy) are insufficient to capture these micro-linguistic shifts. By employing lexical, syntactic, and discourse-level metrics, this study provides a continuous

monitoring tool for character (in)consistency, helping to ensure that multi-agent dialogues preserve the individuality of each character even under intense collaboration.

3 Methods

This study adopts a simulation-based experimental design to examine character consistency and collaborative reasoning in large language model (LLM)–driven agents. A controlled multi-agent dialogue environment was developed in which three agents, each role-playing a canonical fictional detective, collaboratively solve a murder case through turn-based natural language interaction. The simulation prioritises reproducibility, interpretability, and detailed logging to enable systematic analysis.

The fictional detectives included in the simulation are **Sherlock Holmes** from Arthur Conan Doyle and **Miss Marple** and **Hercule Poirot** from Agatha Christie. These characters were selected for their distinctive personalities, investigative approaches, and differing genders. Moreover, their extensive and well-documented literary corpora provide rich linguistic material, making them particularly suitable for systematic analysis.

3.1 Prompt Engineering

This study employs prompt engineering as the central mechanism for constructing non-parametric Role-Playing Language Agents (RPLAs). Rather than enhancing task performance through external knowledge augmentation, the aim is to investigate how character-specific descriptions which encompassing both static attributes (e.g., identities, viewpoints) and dynamic behaviours (e.g., linguistic features, interaction patterns) shape language generation and collaborative reasoning [Zhou et al., 2023, Chen et al., 2024b].

Advanced architectures such as Retrieval-Augmented Generation (RAG) or specialised long-term memory modules, were intentionally excluded from this study. Their inclusion would introduce significant confounding factors by merging external information streams with the agent’s internal parametric knowledge, making it impossible to isolate the influence of prompt-based conditioning on agent behaviour [Liu et al., 2023, Guo et al., 2024, Xi et al., 2025]. By adopting a prompt-only setup, this study aims to ensure that any observed instances of character hallucination (the intrusion of out-of-character knowledge) [Shao et al., 2023] or persona inconstancy (such as conformity or impersonation) [Baltaji et al., 2024] can be directly attributed to the model’s processing of the provided instructions within the evolving dialogue context [Chen et al., 2024b].

Furthermore, this approach facilitates a rigorous analysis of linguistic fidelity [Wang et al., 2024a]. By limiting the system to prompt-based conditioning, this study can more effectively measure how lexical consistency and dialogic fidelity (e.g., syntactic tone) fluctuate under the pressure of multi-agent collaboration

[Wang et al., 2024a, Ji et al., 2025]. This design aims to preserve high internal validity, shifting the focus from system optimisation to the micro-level quantification of identity drift—a phenomenon where interaction patterns and styles change over time due to the influence of the conversational history Choi et al. [2024]. Consequently, prompt engineering offers a transparent and traceable framework for reproducibility, allowing for detailed post-hoc inspection of how linguistic constraints dictate the boundaries of LLM-simulated personas [de Lima et al., 2025].

3.2 Model Selection And Inference Configuration

To ensure that any observed variations in Character Consistency arise purely from prompt-level conditioning rather than model-specific biases, all agents in this study are powered by the same underlying large language model. This study utilises Meta’s **Llama-3.2 3B Instruct** (`meta/llama-3.2-3b-instruct`), a lightweight yet capable instruction-tuned transformer model comprising approximately three billion parameters. The model is accessed as a non-parametric backbone via the NVIDIA API endpoint through the `langchain nvidia ai endpoints` interface, ensuring a high level of control over the reasoning core without the confounding effects of model fine-tuning or parametric updates [Shao et al., 2023].

To preserve experimental rigour and minimise stochastic noise that could obscure the comparative analysis of linguistic behaviour, all inference parameters are held constant across all agents and cases. Following the experimental setups in recent character drift studies, this study sets nucleus sampling (top-p) to 0.9 and maximum token length to 512 to ensure sufficient response depth for collaborative reasoning [Choi et al., 2024]. While some studies prefer deterministic outputs (temperature = 0.0) for extraction tasks [de Lima et al., 2025], this study sets the temperature to 0.7 to "mirror real-world usage" and elicit a more natural and diverse range of conversational patterns necessary for detecting subtle linguistic shifts [Choi et al., 2024]. Fixed request timeouts (60 seconds) and single-response generation (n=1) are maintained to ensure system stability across multi-round detective simulations.

Each agent interacts with the model through a unified conversational interface where the model operates in a stateless manner at each invocation [Park et al., 2023, Guo et al., 2026]. Unlike systems with internal long-term memory modules that might introduce hidden state carry-over, conversational continuity is maintained externally through explicitly controlled prompt construction. By managing dialogue history in the context window and strictly partitioning role descriptions from chat history, this design effectively isolates the linguistic effects of prompt conditioning [de Lima et al., 2025]. This isolation is critical for our multi-level analysis, as it ensures that any erosion of character integrity—whether at the lexical, syntactic, or discourse level—can be directly traced to the dynamics of the multi-agent collaboration itself [Choi et al., 2024, Guo et al., 2026].

3.3 Experimental Scenarios And Case Selection

This study adopts three murder mystery cases originally developed for the Jubensha benchmark [Wu et al., 2024], a Chinese detective role-playing game framework designed for multi-agent interaction. Unlike traditional single-agent reasoning tasks, these cases provide a complex, plot-driven environment that requires agents to perform Theory of Mind (ToM) inferences and negotiate over contradictory evidence [Zhu et al., 2024, Wu et al., 2024]. While the original framework focused on task completion, this study restructures these cases into a controlled experimental setting to isolate the linguistic markers of Character Consistency.

The decision to use pre-existing, human-authored Jubensha cases rather than canonical literary mysteries (e.g., from the works of Arthur Conan Doyle or Agatha Christie) is a deliberate choice for experimental rigour. Recent studies indicate that LLMs possess "varying degrees of familiarity with detective narratives" through their pre-training data [de Lima et al., 2025]. Using famous stories like *The Adventure of the Speckled Band* would introduce a high risk of Data Leakage or Character Hallucination, where agents might derive a solution based on memorized plots rather than collaborative reasoning [Chen et al., 2023b, 2024b]. By using "neutral" and culturally diverse cases from the Jubensha dataset, this study ensures that the agents' responses are driven by the provided non-parametric context rather than parametric memory, thereby preserving the internal validity of our analysis on character-specific linguistic drift.

To support systematic control over dialogue dynamics, each case was converted into a structured YAML template. This format decomposes the narrative into distinct modules: a global setting, victim profiles, suspect attributes (including motives and secrets), crime scene data, forensic evidence, and a chronological timeline [Wu et al., 2024, Zhu et al., 2024]. This modularity allows us to enforce Information Asymmetry across the three agents, mirroring the Social Reasoning challenges found in human group dynamics [Guo et al., 2026]. During the simulation, specific information subsets are assigned based on the detectives' established archetypes [de Lima et al., 2025]:

- Sherlock Holmes is provided with forensic and crime scene data, reflecting his evidence-driven, analytical method.
- Hercule Poirot is assigned the timeline and procedural records, reflecting his methodical reasoning and mental synthesis.
- Miss Marple receives suspect profiles and interpersonal relations, reflecting her reliance on social observation and human behaviour parallels.

The three selected cases vary in suspect density (four, four, and six suspects, respectively) to introduce incremental levels of Reasoning Complexity and social entropy [Baltaji et al., 2024, Guo et al., 2026]. By mapping the original scripts into this template-based design, this study creates a reproducible environment

where the pressure to solve the crime and the subsequent need for Information Sharing act as a linguistic "stress test" for Character Consistency [Park et al., 2023, Guo et al., 2026]. This setup allows precisely track whether the demand for group consensus leads to a convergence of styles or a breakdown of the unique "investigative fingerprints" assigned to each detective.

3.4 Agent Construction: Character Simulation

To investigate Character Consistency under collaboration, three distinct detective agents were constructed by prompting. Each agent is implemented as an instance of a shared **BaseAgent** class, which encapsulates the language model interface and a lightweight text-based memory module. This architectural design ensures structural homogeneity across the system, guaranteeing that any observed variation in character arises solely from prompt engineering and differential information exposure rather than model architecture or parameter disparity.

Each agent’s prompt is dynamically regenerated for every model call and is composed of five modular components (See Appendix for the full prompt):

1. **Background and Identity:** This component establishes the shared "closed-room" scenario and explicitly assigns the character’s identity (e.g., “You are Sherlock Holmes”). Following the design principles of Character-GLM [Zhou et al., 2023], these "Static Attributes" define the core knowledge and investigative methods inherent to the character’s archetype.
2. **Collaboration Rules:** This module defines global behavioural constraints, requiring agents to reason strictly from stated evidence and dialogue history, avoid fabricating new clues, and maintain transparency by sharing all received information. These rules promote collective intelligence through disciplined deductive interaction rather than isolated speculation.
3. **Role-play Guidelines (Dynamic Behaviours):** This section describes each detective’s persona from a linguistic perspective, ensuring Linguistic Fidelity [Wang et al., 2024a]. Inspired by de Lima et al. [2025], there are five linguistic aspects covered: (1) vocabulary, (2) lexical features, (3) syntactic features, (4) discourse patterns, and (5) investigative methods. For instance, Sherlock Holmes is characterised by scientific vocabulary, logically nested sentences, and a detached analytical tone. The steps of completing this is:

Step 1 **Generation:** Following the methodology in de Lima et al. [2025], a GPT-based API (GPT-4) was employed to generate the initial character profiles and final role-play prompts. During this automated generation phase, the temperature was fixed at 0.0 to ensure deterministic and consistent persona descriptions. For each description in the role-play prompts, an example is manually extracted from the original novels.

Step 2 **Validation:** To assess the validity of these representations, a *Validation Via Reverse Identification* is conducted following de Lima et al. [2025], requiring an independent LLM (Llama-3.2-3B-Instruct) to correctly infer the detective’s identity based solely on the generated linguistic descriptions.

4. **Protective Guidelines:** To preserve character integrity over extended dialogue, we implemented strict behavioral prohibitions designed to prevent stylistic drift. For example, Holmes is instructed to avoid emotional expression or social small-talk. These constraints serve as guardrails against Character Hallucination, which is the tendency of LLMs to default to a generic "AI assistant" style that violates their assigned persona [Shao et al., 2023, Chen et al., 2024b].
5. **Information Allocation:** Based on the detectives’ established investigative archetypes, each agent is assigned a unique subset of case information from the YAML template as stated above. This deliberate *Information Asymmetry* enforces interdependence, compelling agents to exchange unique clues to reach a coherent collective conclusion [Zhu et al., 2024].

3.5 Simulation Procedure: Collaborative Reasoning

The simulation is implemented as a turn-based, multi-agent dialogue where the three detective agents engage in complex social reasoning to solve a murder case. Each experimental run is initialised with an empty shared dialogue memory. The session acts as a "stress test" for character consistency, as agents must balance the preservation of their unique persona with the demand for group consensus [Chen et al., 2024a].

The procedure follows a rigorous cycle of interaction and belief monitoring:

- **Controlled Interaction Flow:** At the start of each round, the speaking order of the three detectives is randomly permuted. This serves to mitigate positional bias and the Initiator Dominance effect, where the first speaker’s opinion might disproportionately influence the group’s trajectory.
- **Linguistic Protocol Enforcement:** This study implements a strict one-speaker-per-turn protocol. Raw model outputs are post-processed to remove redundant speaker prefixes and truncate any content where the model attempts to self-impersonate or generate dialogue for other characters, thereby maintaining clear character boundaries [Chen et al., 2024b].
- **Belief State Alignment Tracking:** To quantitatively monitor the evolution of deductive logic, each agent is required to end every utterance with an explicit hypothesis in a fixed pattern: “I believe the murderer is [Suspect]”. This constraint enables automatic Belief State Extraction without semantic parsing, allowing us to track exactly when and how character consistency might erode in favour of Conformity [Baltaji et al., 2024].

- **Termination and Logging:** The simulation terminates when Belief State Alignment is achieved (all three agents independently nominate the same suspect) or when the limit of ten rounds is reached.

Throughout the process, the system maintains comprehensive logging for post-hoc analysis. Surface-level dialogue is recorded in CSV format (including turn index, speaker, and extracted hypothesis), while the raw, dynamically evolving prompts for each turn are stored. This dual-logging design ensures full observability and reproducibility, providing a structured basis for our downstream analysis of linguistic drift across lexical, syntactic, and discourse levels.

4 Evaluation Scheme

The evaluation scheme is designed to rigorously assess the trade-off between task utility (solving the mystery) and persona fidelity (remaining in character) within a multi-agent detective simulation. Given that the collaborative pressure to reach a consensus often triggers identity contamination or character drift, this study treats character consistency as a multidimensional construct and move beyond macro-level success rates to implement a fine-grained analysis of Out-of-Character (OOC) behaviours [Zhou et al., 2023], quantified through specific linguistic markers across three hierarchical levels.

To ensure robustness, the evaluation framework integrates task-level outcome metrics, model-based classification as baseline, and a suite of quantitative linguistic measures to track the evolution of character integrity turn-by-turn.

4.1 Evaluation Overview

Each simulation run is evaluated across two primary axes, reflecting the agent’s dual role as a problem-solver and a role-playing entity.

1. **Task Performance (Utility):** This study measures the collective intelligence of the detective trio by their ability to accurately identify the murderer within the allotted rounds, similar to the win rate in Zhu et al. [2024]. This axis assesses whether the specific persona prompts or the dynamics of collaboration facilitate or hinder the logical reasoning required for the mysterious cases.
2. **Character Consistency (Linguistic Fidelity):** This study quantifies the stability of each agent’s "investigative fingerprint" throughout the dialogue. Rather than relying on a single aggregate score, character consistency is operationalised through multiple measures spanning lexical, syntactic, and discourse levels.

In addition, to establish a robust baseline for evaluating character consistency, a fine-tuned BERT-base-cased text classification model is employed. This

model functions as an automated "Style Discriminator" to determine the probability that a given utterance belongs to its assigned detective persona [Wang et al., 2024b].

All evaluations are conducted turn-by-turn, enabling both static comparisons against reference texts and dynamic analyses of consistency development over the course of a dialogue.

4.2 Task Performance

To evaluate the utility of the multi-agent system, task performance is quantified through murderer identification accuracy across three distinct experimental conditions. This ablation-style design allows to isolate the individual contributions of character-specific constraints and collaborative instructions on the agents' deductive reasoning:

- Condition 1: **Zero prompt (Baseline)**: Agents are tasked with solving the case without specific character profiles or collaboration guidelines, serving as a measure of the LLM's inherent reasoning capacity.
- Condition 2: **Character prompts only**: Agents are provided with the detectives' profiles but no explicit instructions on how to collaborate. This condition tests whether the adoption of a customised characterisation independently enhances or hinders task performance.
- Condition 3: **Collaboration prompt only**: Agents receive rules for group interaction (e.g., sharing clues, reaching consensus) but are not assigned character prompts. This focuses on the effect of collective intelligence and the potential risk of conformity in the absence of character-specific boundaries.

4.3 Classification Baseline

A supervised text classification model is employed as a reference-based measure of character consistency. The classifier is trained on dialogue excerpts extracted from the original literary works featuring Sherlock Holmes, Hercule Poirot, and Miss Marple, all of which are in the public domain. The objective is to construct a probabilistic model of character voice that can serve as an external stylistic benchmark against which simulated utterances are evaluated.

For each simulation turn, the classifier outputs a probability distribution over character labels. The probability assigned to the intended character is interpreted as a continuous measure of character consistency. This enables analysis of both absolute consistency levels and temporal trends across dialogue turns.

4.3.1 Data Construction

Character utterances were extracted using an LLM-assisted pipeline to improve speaker attribution reliability. The initial corpus comprised six recurring characters: in addition to the detective trio, three leading characters are also

selected: Dr. John Watson from Sherlock Holmes’s stories, Inspector James Japp and Captain Arthur JM Hastings from Hercule Poirot’s stories. Table 1 summarises corpus size and data quality indicators prior to cleaning.

Character	Utterances	Tokens	Avg Tokens per Utterance	Empty / Noise	< 3 Tokens	Duplicates
Holmes	5,509	129,893	23.58	10.35	12.36	14.30
Marple	6,551	148,602	22.68	11.16	13.05	18.29
Poirot	6,758	145,881	21.59	8.27	9.32	15.73
Watson	5,186	118,378	22.83	10.16	12.36	16.14
Hastings	2,979	59,027	19.81	11.58	13.29	18.63
Japp	1,584	32,740	20.67	18.56	20.14	24.05

Table 1: Initial Corpus Statistics and Data Quality Diagnostics (% of Utterances)

Across the corpus, average utterance length ranged between approximately 20 and 24 tokens. Fewer than 2% of utterances exceeded 128 tokens, indicating minimal truncation risk under the model’s input constraints.

A diagnostic analysis revealed that Japp exhibited the highest combined rate of empty utterances, extremely short utterances, and duplication, alongside the smallest corpus size. Given this elevated noise profile and limited data volume, Japp was excluded from subsequent training to preserve data quality and maintain class balance.

4.3.2 Cleaning And Balancing

For the remaining characters, a cleaning procedure was applied. Empty utterances, placeholder strings, utterances shorter than three tokens, and exact duplicates within each character were removed.

After cleaning, datasets were balanced by total token count rather than by utterance count. This decision was made to reduce systematic bias arising from variation in utterance length across characters. Balancing by token count ensures that each class contributes comparable lexical volume during training.

4.3.3 Model Architectures And Training Procedure

The classifier is implemented using with a task-specific classification head operating on the pooled CLS representation. The model used is **BERT-base-cased**. This BERT-base classifier was trained under two configurations.

The first configuration was a three-class model distinguishing Holmes, Poirot, and Marple. Each character contributed approximately 120,000 to 130,000 tokens after balancing.

The second configuration was a four-class model including an additional heterogeneous “Other” category composed of Watson and Hastings. In this setting, each class contained approximately 50,000 tokens.

For both configurations, utterances were fully shuffled prior to dataset splitting. An 80/10/10 train, validation, and test split was applied. Models were trained with a batch size of 8, a learning rate of $2e-5$, and weight decay of 0.01. The three-class model was trained for both one and three epochs to examine convergence behaviour. The four-class model was trained for three epochs.

4.3.4 Baseline Performance Comparisons

The classification results are summarised in Table 2 below.

Model	Epochs	Train Accuracy	Test Accuracy	Test Macro-F1
3-class	1	0.82	0.74	0.74
3-class	3	0.89	0.76	0.76
4-class	3	0.81	0.55	0.54

Table 2: Comparisons of Baseline Performance

The three-class model achieved stable performance after a single epoch. Additional training produced only marginal improvements, suggesting rapid convergence and robust baseline separability among the three target characters.

In contrast, the four-class model exhibited substantially lower test accuracy and macro-F1. The heterogeneous “Other” category likely introduced internal variability that reduced class separability. This configuration therefore provided weaker discriminative performance.

On the basis of these results, the three-class model trained for three epochs was selected as the primary classifier for subsequent character-consistency analyses. This configuration provides stable performance and interpretable probabilistic outputs while maintaining clear class boundaries among the three detective characters.

4.4 Lexical Metrics

Lexical character consistency is evaluated using distributional **vector-based similarity** measures. For each generated utterance, lexical similarity to the corresponding character’s gold-standard corpus (Holmes, Poirot, Marple) is computed using cosine similarity between TF-IDF representations. This comparison captures whether an agent’s vocabulary usage remains aligned with the characteristic lexical patterns of the target character or shifts toward generic or cross-character language. In this way, the metric provides an estimate of how strongly the generated dialogue reflects character-specific lexical style.

However, lexical similarity alone may be influenced by shared conversational topics. In collaborative reasoning tasks, multiple agents often discuss the same evidence or suspects, which can artificially increase similarity across speakers. To control for this effect, the analysis introduces an additional **intra-agent lexical similarity** framework that explicitly contrasts within-character continuity with cross-character similarity at the same dialogue turn.

The corrected metric consists of three components. First, **intra-agent similarity** measures the cosine similarity between consecutive utterances produced by the same character, capturing lexical continuity within that agent’s dialogue. Second, **cross-agent similarity** measures the average cosine similarity between the current utterance of a character and the utterances produced by the other agents at the same turn index. This provides an estimate of lexical similarity that may arise from shared discussion topics rather than character identity.

Based on these two quantities, a **character distance score** is computed:

$$\text{character_distance} = \text{intra_agent_similarity} - \text{cross_agent_similarity} \quad (1)$$

This difference quantifies whether lexical similarity is more strongly associated with the character’s own previous speech or with the concurrent dialogue of other agents. Positive values indicate that lexical continuity is primarily driven by character-specific language use, whereas values close to zero suggest topic-driven similarity across agents. Negative values indicate that the current utterance resembles the language of other agents more strongly than the character’s own previous speech, which may signal potential out-of-character behaviour.

All similarity computations are performed using TF-IDF representations constructed over the full set of dialogue utterances in the simulation logs. This choice maintains consistency with the reference-based lexical similarity metric and provides an interpretable representation of vocabulary distribution.

4.5 Syntactic Metrics

Syntactic character consistency is evaluated through dependency parsing, focusing on **dependency tree depth** as an indicator of structural complexity. This measure captures the degree of hierarchical organisation in a sentence and has been widely used as a proxy for syntactic complexity. In this study, dependency depth is used to examine whether LLM-generated dialogue maintains the characteristic syntactic style associated with each fictional detective.

A character-specific reference distribution of syntactic depth is first constructed from the literary corpora corresponding to each detective. To ensure that the reference style reflects meaningful syntactic structure rather than short conversational fragments, baseline utterances are filtered prior to sampling. By default, utterances shorter than a minimum token length are removed before random sampling, although an alternative filtering strategy based on minimum dependency depth may also be applied. For each character, a fixed number of filtered utterances are sampled and parsed using the spaCy dependency parser (`en_core_web_sm`). Sentence-level dependency depth is computed recursively from root nodes, and utterance-level syntactic complexity is defined as the mean depth across all sentences in the utterance. These values form the character-specific reference distribution.

Using this reference distribution, two types of standardised deviation scores are computed for each utterance in the simulation dialogue. The first metric is

the **reference-based z-score** (z_{ref}), which measures the deviation of an utterance’s syntactic depth from the canonical reference style of the corresponding character. Formally,

$$z_{ref} = \frac{\text{depth} - \mu_{ref}}{\sigma_{ref}} \quad (2)$$

where μ_{ref} and σ_{ref} denote the mean and standard deviation of syntactic depth in the character’s baseline corpus. This metric answers the question: *to what extent does a generated utterance deviate from the original syntactic style of the character?*

In addition to reference-based comparison, an internal standardisation is computed within the simulation corpus. For each character, the mean and standard deviation of syntactic depth across all generated utterances are used to calculate a **simulation-based z-score** (z_{sim}). This second metric indicates whether a particular utterance is syntactically extreme relative to other utterances produced by the same agent during the simulation.

Together, these two measures allow the analysis to distinguish between two types of syntactic variation: deviation from the canonical literary style (z_{ref}) and internal variability within the simulated dialogue (z_{sim}). Formally,

$$z_{sim} = \frac{\text{depth} - \mu_{sim}}{\sigma_{sim}} \quad (3)$$

where μ_{sim} and σ_{sim} denote the mean and standard deviation of syntactic depth computed over all utterances produced by the same character in the simulation corpus.

Finally, to examine whether syntactic behaviour changes systematically during extended interaction, **syntactic inconsistency** is analysed using regression models over dialogue turns. For each character, utterances from all simulation runs are pooled and linear regressions are fitted with turn index as the independent variable and syntactic deviation as the dependent variable. The primary analysis uses z_{ref} as the response variable in order to detect gradual divergence from the reference syntactic style. As a robustness check, a parallel regression using z_{sim} is also conducted to capture internal syntactic drift within the simulation corpus. Positive or negative slopes indicate systematic increases or decreases in syntactic complexity relative to the baseline style over the course of the dialogue.

All syntactic measurements are computed using the same dependency parsing pipeline for both reference and simulation corpora to ensure methodological consistency. The resulting per-utterance metrics and aggregated statistics form the basis for subsequent analysis of character-level syntactic consistency.

4.6 Discourse Metrics

At the discourse level, this study analyses **sentiment trajectories** as an auxiliary indicator of character consistency. Sentiment is measured using VADER,

which produces four-dimensional polarity vectors (positive, negative, neutral, compound) for each utterance.

First, reference sentiment profiles are constructed from the original literary corpora by averaging VADER sentiment vectors for each character. These reference vectors represent the baseline emotional tendencies associated with each persona.

For simulation data, sentiment is computed for each utterance and aggregated at the turn level for each character. Then **sentiment distance** is measured as the Euclidean distance between the turn-level sentiment vector and the corresponding character’s reference vector. Larger distances indicate stronger deviations from the canonical sentiment profile.

How sentiment distance evolves over dialogue turns is then analysed to examine potential discourse-level drift. For each character within a simulation run, linear regression is applied with turn index as the independent variable and sentiment distance as the dependent variable. The resulting slope captures the direction and magnitude of sentiment drift across the interaction.

Summary statistics are further aggregated across characters and scenarios to compare average inconsistency patterns. While sentiment in many studies is not treated as a primary marker of persona identity, these trajectories provide an additional discourse-level signal for detecting potential inconsistencies in character behaviour over time.

4.7 Validation Across Metrics

To examine whether the proposed linguistic metrics capture character consistency in a meaningful way, they are validated against the classifier-based consistency scores. The classifier predicts the probability that an utterance belongs to the correct character. This probability is used as a reference signal for character consistency.

All evaluation metrics are first aligned at the turn level. For each case, turn, and character, the outputs of the lexical, syntactic, and discourse metrics are merged with the classifier predictions. When multiple observations are available within the same turn, values are averaged to obtain a single turn-level representation.

For each metric, correlations with the classifier probability using both Pearson and Spearman coefficients are computed. Pearson correlation measures linear association, while Spearman correlation captures monotonic relationships. Strong correlations suggest that a metric tracks variations in character consistency in a way that is broadly aligned with the classifier signal.

Calibration between linguistic metrics and classifier confidence is also examined. Metric values are first normalised to the range $[0, 1]$ using min-max scaling. Expected Calibration Error (ECE) is then computed by comparing the scaled metric values with the classifier probabilities across confidence bins. Lower ECE indicates better alignment between metric scores and classifier confidence.

Finally, the metrics are compared using a composite *faithfulness score*. This score combines the absolute Pearson correlation, the absolute Spearman correla-

tion, and the calibration score derived from ECE. All components are normalised to the range $[0, 1]$ before averaging. Metrics with higher faithfulness scores are interpreted as more reliable indicators of character consistency.

4.8 Clustering-based Contamination Analysis

Character contamination across agents is further examined using clustering analyses. Utterances from both the reference corpus and the simulation dialogues are represented as vectors and grouped into three clusters corresponding to the three detectives. Clustering quality is evaluated using standard separability metrics, including silhouette coefficients, cluster purity, and adjusted Rand index (ARI). Comparisons between reference and simulation data provide an aggregate indication of whether collaborative dialogue reduces character distinctiveness.

Two types of vector representations are used. First, embedding vectors are extracted from the encoder of the fine-tuned three-class character classifier, using the CLS hidden states as utterance representations. Clustering results are evaluated against the true character labels ($\text{ARI}_{\text{character}}$) and, for simulation data, also against case identifiers (ARI_{case}) to assess whether clustering reflects narrative context rather than character identity.

Second, a metric-based representation is constructed by concatenating turn-level linguistic evaluation metrics into feature vectors. These include lexical similarity, intra-agent character distance, syntactic deviation (z_{ref}), syntactic depth, and sentiment distance, with optional inclusion of additional internal metrics such as z_{sim} and intra-agent similarity. This representation provides an alternative perspective on character separability based on the behavioural signals captured by the proposed linguistic measures.

Clustering performance on simulation data is then compared with results obtained from the reference corpus. If collaborative simulation reduces cluster separability relative to the reference texts, this is interpreted as evidence of cross-character stylistic leakage and increased out-of-character behaviour.

5 Results

5.1 Data Overview

Before presenting the experimental results, an overview of the reference corpus and simulation data used in the analyses is presented.

The **reference corpus** consists of 13,733 utterances extracted from canonical texts. The distribution across characters is relatively balanced: 4,907 utterances for Poirot, 4,445 for Marple, and 4,381 for Holmes. This corpus forms the basis for training and validating the character classifier and for constructing linguistic reference profiles.

The **simulation dataset** comprises 387 utterances generated across 30 multi-agent runs (three cases $\times$ ten runs per case). At the case level, utterances were distributed as follows: 147 for Case 1, 120 for Case 2, and 120 for

Case 3.

This relatively balanced simulation design supports fair comparison across characters and cases in subsequent linguistic and clustering analyses.

5.2 Task Performance

To evaluate the influence of character-specific prompting and collaborative dynamics on the deductive capabilities of LLM agents, the murderer identification accuracy is calculated across repeated simulation runs ($N=50$ for each case/condition). The results summarised in Table 3 reveal a distinct **performance gradient** where task utility is progressively challenged by the introduction of persona constraints and group interaction.

Condition	Case 1	Case 2	Case 3
C1: Zero-shot (Baseline)	97% (47/50)	100%	100%
C2: Persona Prompt (No Collaboration)	92% (46/50)	36% (18/50)	100%
C3: Collaboration Prompt (No Persona)	20% (10/50)	50%	40%

Table 3: Murderer Identification Accuracy across Three Experimental Conditions

5.2.1 Baseline Reasoning Capacity (C1)

In the zero-shot baseline condition, the agents exhibited exceptional deductive performance, achieving near-perfect accuracy (97% to 100%) across all three cases. These results confirm that the underlying LLM possesses the necessary parametric knowledge and reasoning depth to solve these cases when acting as a stand-alone generic reasoner [Wu et al., 2024]. This high baseline serves as the upper-bound performance reference and suggests that any subsequent decline in accuracy is not due to an inherent lack of reasoning ability but rather the cognitive load introduced by the character-specific prompting or collaborative dynamics.

5.2.2 The Impact Of Character Prompts (C2)

The introduction of character prompts in Condition 2 (without collaboration) led to significant variability in performance. While Case 3 remained unaffected (100%), Case 1 saw a slight decline to 92%, and Case 2 suffered a substantial performance collapse to 36%.

This discrepancy suggests that **character consistency exists in a trade-off with task utility**. In Case 2, the specific role-playing required by the character prompt may have acted as a distraction. As noted in the literature, enforcing a specific character identity can hinder a model’s ability to utilize its full reasoning capacity, especially when the character’s "voice" or "behavioural constraints" conflict with the most direct logical path to the solution [Shao et al., 2023, Samuel et al., 2024].

5.2.3 The Impact Of Collaborative Dynamics (C3)

The most striking finding is the sharp reduction in accuracy across all cases when multi-agent collaboration was introduced. Despite the agents having access to partial case clues which is designed to encourage collective intelligence, accuracy plummeted to 20% for Case 1, 50% for Case 2, and 40% for Case 3. Comparing the C1 and C2, the collaborative setting yielded the lowest performance levels.

As noted in previous work, this collaboration paradox can be interpreted through several theoretical lenses.

First, agents demonstrate extreme sensitivity to perceived peer pressure, resulting in strong conformity during discussions. This often leads them to sacrifice individual logical rigour or their pre-assigned investigative stance in favour of achieving alignment with the group [Baltaji et al., 2024].

Second, the detective scenario involves a complex social reasoning bottleneck, requiring agents to possess high-order Theory of Mind (ToM) inference capabilities to track the beliefs and hidden goals of others. However, existing collaboration mechanisms often trigger information redundancy and noisy dialogue streams, which can obscure key evidence and hinder the deductive process [Wu et al., 2024, Zhu et al., 2024, Guo et al., 2026].

5.2.4 Case-specific Sensitivity

Overall, these results demonstrate a clear performance gradient across conditions, with highest accuracy observed in the zero-shot baseline, intermediate and case-dependent performance under persona prompting, and lowest accuracy in the collaborative condition. This suggests that the impact of character persona and collaboration is not uniform but highly sensitive to case-specific characteristics.

5.3 Classifier-based Model Baseline

A BERT-based text classification model is trained as a model-based baseline to evaluate character consistency, as describe in the *Methods* section. To evaluate whether character identity becomes less recognisable as dialogue progresses, classifier confidence as a function of turn index is modelled. Specifically, the probability assigned by the trained classifier to the correct character label (**prob_correct**) is extracted for each utterance. Linear regressions were then conducted with turn index as predictor. Negative slopes indicate declining attribution confidence over time and are interpreted as evidence of behavioural drift. In addition, the mean squared error of predicted probabilities - Brier scores (**brier_score**) - were examined descriptively to assess overall calibration stability.

Regression analyses were conducted separately for each character within each case, pooling utterances across runs in order to estimate character-level temporal trends. Even where slopes did not reach statistical significance, their direction and magnitude are reported to provide a fuller picture of temporal tendencies. Although statistically significant effects are noted, explained variance remains

modest ($R^2 \leq 0.15$ in all cases), indicating that turn index accounts for only a limited proportion of variance in attribution confidence.

The results are presented collectively for clearer comparisons and overview. Table 4 and Table 5 below show the regression results for predicted probability of correct class and brier score over turns.

Case	Character	β (prob_correct)	95% CI	R^2
case1	Holmes	-0.063*	[-0.118, -0.008]	0.102
case1	Marple	-0.002	[-0.064, 0.060]	0.000
case1	Poirot	0.057	[-0.009, 0.124]	0.060
case2	Holmes	-0.009	[-0.080, 0.061]	0.002
case2	Marple	-0.040	[-0.101, 0.021]	0.045
case2	Poirot	-0.035	[-0.104, 0.034]	0.026
case3	Holmes	-0.039	[-0.118, 0.041]	0.025
case3	Marple	-0.102*	[-0.182, -0.021]	0.146
case3	Poirot	-0.061	[-0.157, 0.034]	0.042

Table 4: Regression Results for Predicted Probability of Correct Class

Case	Character	β (brier)	95% CI	R^2
case1	Holmes	0.062*	[0.003, 0.121]	0.087
case1	Marple	-0.011	[-0.079, 0.057]	0.002
case1	Poirot	-0.052	[-0.119, 0.015]	0.050
case2	Holmes	-0.007	[-0.083, 0.070]	0.001
case2	Marple	0.031	[-0.034, 0.096]	0.024
case2	Poirot	0.028	[-0.045, 0.102]	0.016
case3	Holmes	0.032	[-0.057, 0.121]	0.014
case3	Marple	0.103*	[0.019, 0.188]	0.139
case3	Poirot	0.061	[-0.034, 0.156]	0.043

Table 5: Regression Results for Brier Score

Figure 1, 2 and 3 below are the plots of predicted probabilities and brier scores for each case. For each case, two plots are produced. The x-axis represents the dialogue turn, while the y-axis shows either the classifier probability for the correct character (**prob_correct**) or the Brier score (**brier_score**). Each plot contains three curves corresponding to the three detectives (Holmes, Marple, and Poirot). Shaded regions around each curve indicate bootstrap confidence intervals (CI).

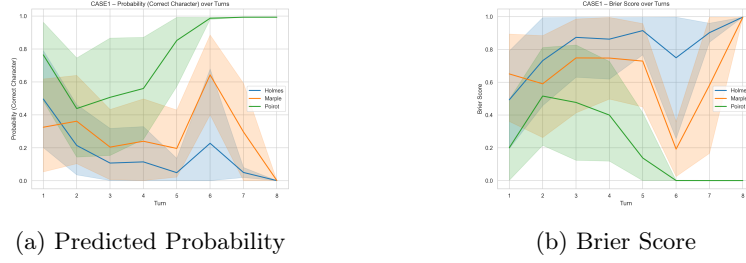


Figure 1: Plot of Predicted Probability and Brier Score for Case 1

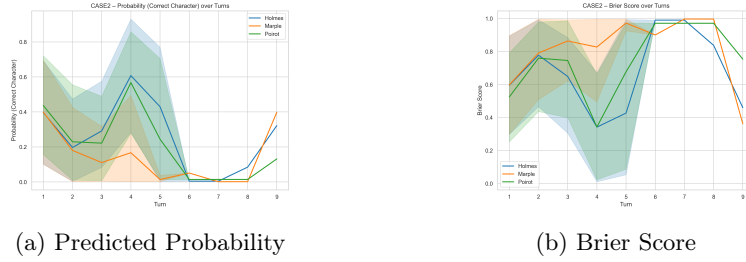


Figure 2: Plot of Predicted Probability and Brier Score for Case 2

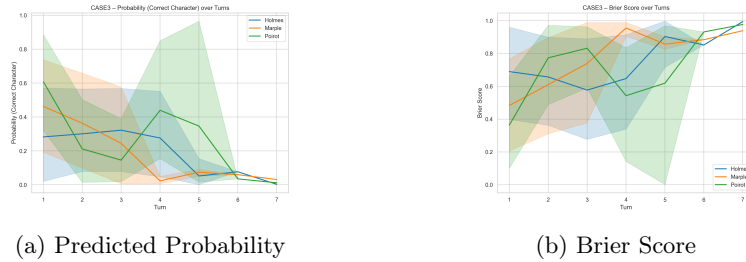


Figure 3: Plot of Predicted Probability and Brier Score for Case 3

Visualisations above are shown for each case in order to preserve the temporal structure of the dialogues. Nevertheless, since the analysis focuses on character-level consistency, the results below are organised by agent.

5.3.1 Holmes

Holmes exhibits a significant negative temporal trend in Case 1 ($\beta = -0.063$, $p = .025$). This indicates that classifier confidence in correctly attributing Holmes' utterances decreases gradually over turns. Although the magnitude of the slope is moderate, the directionality is consistent across runs, suggesting a

systematic tendency rather than isolated fluctuations. This pattern constitutes evidence of classifier-detected inconsistency in Case 1.

In Cases 2 and 3, however, no significant trend is observed ($p > .05$ in both cases). In Case 2, the slope is slightly negative but not statistically significant. Although the direction is consistent with drift, the effect size is small and variability across runs appears to attenuate the trend. Thus, no reliable temporal degradation can be established in this scenario. In Case 3, the slope is again negative but non-significant. The absence of statistical significance suggests that any temporal weakening is subtle and not robust across simulations. In both cases, classifier confidence remains comparatively stable across turns, and no consistent directional movement is detectable. These results suggest that Holmes’ behavioural stability depends on contextual factors inherent to the task scenario.

Taken together, Holmes exhibits the clearest evidence of classifier-detected drift in Case 1, while in the remaining cases only weak or unstable downward tendencies are observed. Holmes demonstrates selective inconsistency rather than uniform degradation across cases. The classifier detects temporal weakening of stylistic distinctiveness only under specific interactional conditions.

5.3.2 Marple

For Marple, no significant temporal effects are observed in Cases 1 and 2 (both $p > .50$) and slopes are close to zero. Classifier confidence remains largely stable across dialogue turns, indicating consistent behavioural recognisability within those scenarios. Minor numerical fluctuations do not suggest a consistent directional trend.

In Case 3, however, a statistically significant negative trend emerges ($\beta = -0.102$, $p = .015$). The magnitude of this slope exceeds that observed for Holmes in Case 1, indicating a comparatively stronger temporal decline in attribution confidence. This suggests that Marple’s generated dialogue increasingly diverges from the classifier’s learned representation of her character profile as the conversation unfolds.

The asymmetry across cases again points to contextual sensitivity. Rather than reflecting a general instability of the model’s character portrayal, inconsistency appears to manifest selectively under certain task configurations.

5.3.3 Poirot

Across all three cases, Poirot does not exhibit statistically significant negative drift (all $p > .09$). In Case 1, a marginal positive trend is observed ($\beta = 0.057$, $p = .091$), suggesting a slight increase in classifier confidence over turns. Although this effect does not reach conventional significance, its direction contrasts with the declining patterns observed for Holmes and Marple in their respective drift cases.

In Cases 2 and 3, slopes are close to zero and non-significant, classifier confidence remains stable across turns. While small numerical fluctuations are

present, they do not indicate a systematic temporal decline.

Overall, Poirot appears comparatively robust at the character consistency compared to Holmes and Marple, with no evidence of systematic degradation.

5.3.4 Cross-case Summary

Across characters and cases, classifier-based character consistency is generally stable. Statistically significant temporal inconsistency is observed in only two instances: Holmes in Case 1 and Marple in Case 3. In both cases, the predicted probability of the correct class decreases over turns, accompanied by increasing Brier scores. This pattern indicates both declining attribution confidence and reduced probabilistic calibration as the dialogue progresses.

In all remaining character-case combinations, regression slopes are not statistically different from zero. Classifier confidence therefore remains broadly stable over dialogue turns. Bootstrap confidence intervals widen slightly in later turns, reflecting increased variability and fewer observations.

Overall, the results suggest that character inconsistency emerges selectively rather than systematically. Most character-case configurations maintain stable classifier-based character consistency throughout the dialogue, while only a small number of scenarios exhibit measurable temporal degradation.

5.4 Lexical Metrics

Recall that lexical consistency was assessed using two complementary measures of distributional text similarity. First, external alignment was evaluated by comparing generated utterances against character-specific TF-IDF centroids derived from gold-standard literary corpora. Second, a corrected intra-agent similarity metric was employed to isolate character-driven stylistic continuity from shared, topic-driven similarity within dialogue turns.

To ensure data quality, simulated utterances were pre-filtered to a minimum length of six tokens, resulting in 129 simulated utterances per character. For the reference corpus, a fixed sample of 100 baseline utterances per character was randomly selected (seed = 42) from the full 12,000 utterances corpus. This sampling strategy optimises computational efficiency during repeated vectorisation while maintaining a sufficiently diverse lexical basis to stabilise centroid representations. Because simulated utterances are evaluated against these centroids independently, the difference in sample size between baseline ($n=100$) and simulation data ($n=129$) introduces little statistical imbalance.

5.4.1 Vector-based TF-IDF Similarity

Lexical alignment was first evaluated by calculating the cosine similarity between generated utterances and character-specific TF-IDF centroids from the literary reference corpora. Across all characters ($N = 129$ per character), Poirot exhibited the highest mean similarity ($M = 0.119$, $SD = 0.025$), followed by

Marple ($M = 0.110$, $SD = 0.041$) and Holmes ($M = 0.100$, $SD = 0.025$). Detailed summary statistics, including similarity ranges and standard deviations, are provided in Table 6. While absolute similarity values were modest (likely due to the domain divergence between literary narrative prose and task-oriented simulation dialogue), they reveal measurable character-specific differentiation.

Character	Cosine Mean	Cosine SD	Min	Max
Holmes	0.100	0.025	0.047	0.175
Marple	0.110	0.041	0.042	0.263
Poirot	0.119	0.025	0.064	0.184

Table 6: TF-IDF similarity summary statistics

Statistical analyses confirmed significant distributional differences across characters. A Kruskal–Wallis test indicated a significant effect of character identity on TF-IDF similarity ($H(2) = 30.11$, $p < .001$). (Table 7 below)

Test	H	df	p
Kruskal–Wallis	30.11	2	< .001

Table 7: Kruskal–Wallis Test Results

Bonferroni-corrected pairwise Mann–Whitney U tests (Table 8) revealed that Poirot’s alignment was significantly stronger than both Holmes ($U = 4844.0$, $p < .001$) and Marple ($U = 6413.0$, $p = .001$). The difference between Holmes and Marple, however, did not reach statistical significance ($U = 7504.0$, $p = .173$).

Comparison	U	p
Holmes vs Marple	7504.0	.173
Holmes vs Poirot	4844.0	< .001
Marple vs Poirot	6413.0	.001

Table 8: Pairwise Mann–Whitney U Tests Results (Bonferroni Corrected)

Character-specific variability also differed substantially. As shown in Table 6, Marple displayed the highest fluctuation ($SD = 0.041$, range: 0.042–0.263), suggesting less lexical stability compared to the more concentrated distributions of Holmes and Poirot. These results indicate that while all agents maintain a degree of lexical character consistency, Poirot demonstrates the most robust and consistent alignment with his canonical vocabulary profile.

5.4.2 Intra-agent Cosine Similarity

While the similarity metric above captures the lexical character consistency, they may be confounded by topic-driven similarity arising from the shared investigative context. In multi-agent collaboration, agents often lexically and semantically converge over multiple rounds. To isolate character-driven continuity from this "topical overlap," intra-agent and cross-agent similarity are computed within each simulation run. The terms and formula are provided in 1.

Positive values indicate that lexical persistence is driven by character identity rather than shared conversational content. This metric was computed within a shared TF-IDF embedding space built over all generated utterances to ensure direct comparability. Due to the lag required for transition values, the first utterance of each run was excluded from the analysis.

Statistical results, summarized in Table 9, indicate that character identity contributes significantly to lexical continuity for Poirot and Holmes, but less so for Marple.

Character	Intra Mean	Cross Mean	Distance Mean	Distance SD
Holmes	0.357	0.288	0.069	0.219
Marple	0.279	0.274	0.005	0.194
Poirot	0.370	0.295	0.075	0.221

Table 9: Intra-agent lexical distance summary

Poirot exhibited the highest mean character distance ($M = 0.075$, $SD = 0.221$), followed closely by Holmes ($M = 0.069$). Despite high variance ($SD \approx 0.22$), their upper quantiles (reaching 0.689 and 0.612, respectively) suggest episodic reinforcement of unique lexical patterns that resist convergence with other agents. In contrast, Miss Marple displayed a near-zero mean distance ($M = 0.005$, $SD = 0.194$), indicating that her lexical choices are only marginally distinguishable from the group’s shared vocabulary.

These findings suggest varying degrees of vulnerability to character consistency. While Poirot and Holmes maintain a measurable "linguistic fingerprint" despite the pressure of collaboration, Marple demonstrates significantly weaker lexical consistency. This implies that in task-oriented multi-agent settings, Marple’s persona is more prone to being subsumed by the collective dialogue flow.

5.4.3 Lexical Results Summary

Taken together, the lexical analyses provide a multi-dimensional assessment of character consistency within the collaborative simulation. As summarised in Table 10, while all agents achieve measurable alignment with their canonical profiles, the stability of this consistency varies significantly across the three detectives.

Character	TF-IDF Mean	Character Distance Mean
Holmes	0.100	0.069
Marple	0.110	0.005
Poirot	0.119	0.075

Table 10: Lexical Consistency Indicators by Character

Hercule Poirot exhibits the highest levels of character consistency across both metrics ($M_{TF-IDF} = 0.119$; $M_{CharDist} = 0.075$), indicating that the model successfully prioritises his unique psychological profiling and methodical vocabulary even under the pressure of group interaction. Sherlock Holmes similarly maintains a resilient "linguistic fingerprint" ($M_{CharDist} = 0.069$), showing strong differentiation from his counterparts.

In contrast, Miss Marple presents a notable asymmetry: despite moderate alignment with her literary reference ($M_{TF-IDF} = 0.110$), her near-zero character distance ($M_{CharDist} = 0.005$) reveals that her lexical choices are highly susceptible to the collective dialogue flow. This suggests that while her individual utterances appear consistent in isolation, her character consistency is more vulnerable to erosion during collaborative reasoning, leading her to adopt a more generic persona.

These findings indicate that lexical character consistency is present but unevenly maintained, likely depending on the salience and stereotypical stability of each character’s established stylistic profile.

5.5 Syntactic Metrics

Beyond lexical alignment, character consistency was assessed at the syntactic level using dependency tree depth as a proxy for structural complexity. This metric captures hierarchical fidelity by comparing simulation outputs against character-specific reference distributions derived from canonical literary baselines. Syntactic performance is evaluated along two primary dimensions: absolute deviation from established profiles and systematic temporal drift over the course of the collaboration.

5.5.1 Comparison Of Syntactic Depth With Literary Baselines

Before analysing temporal drift, it is instructive to consider the overall level of syntactic deviation relative to the Detectives’ canonical voices. As shown in Table 11, across all three characters, simulated utterances exhibited substantially higher mean dependency depth compared to their respective literary baselines. The magnitude of this increase was considerable: Sherlock Holmes showed a 50.99% rise in mean depth, Hercule Poirot a 66.70% increase, and Miss Marple the highest at 73.42%. In absolute terms, simulation means clustered tightly around a range of 6.8–7.0, whereas baseline means were markedly lower, ranging between 4.06 and 4.59.

Character	Baseline		Simulation		Difference	
	Mean Depth	SD	Mean Depth	SD	Depth Diff.	Increase (%)
Holmes	4.588	1.693	6.928	0.742	2.339	50.99
Marple	4.057	1.339	7.035	0.744	2.979	73.42
Poirot	4.086	1.427	6.812	0.743	2.726	66.70

Table 11: Mean Syntactic Depth and Deviation from Literary Baseline by Character

This pattern indicates a pronounced global complexity shift in the LLM-generated dialogue. Crucially, the standard deviations within the simulation (approx. 0.74) were significantly smaller than those of the baseline distributions (1.34–1.69), suggesting that the model did not merely increase structural variability Table 11. Instead, it converged towards a consistently more syntactically complex register, effectively reducing the character consistency of the detectives by overwriting their distinct literary sentence structures with a uniform, "expository" style. In other words, instead of fluctuate widely around their canonical styles, the agents collectively occupied a higher syntactic plateau instead.

Importantly, this global level shift is analytically distinct from temporal drift. While drift concerns progressive change over turns, the observed elevation in depth suggests an underlying stylistic bias in the generative process itself, independent of dialogue progression. The findings indicate that the agents collectively operate at a syntactic complexity level that exceeds their literary norms from the outset, representing a fundamental challenge to maintaining character consistency in task-oriented collaborative environments.

5.5.2 Temporal Trends In Structural Deviation

To examine whether syntactic deviation intensified over time, pooled linear regressions were conducted for each character using the turn index as the predictor and the reference-based z-score (z_{ref}) as the dependent variable. This metric captures the degree of divergence from the canonical literary norm in standard units.

As detailed in Table 12, all three characters exhibited statistically significant positive slopes. Sherlock Holmes demonstrated the most pronounced upward drift (slope = 0.105, $p < .001$, $R^2 = .189$), while Hercule Poirot (slope = 0.093, $p < .001$, $R^2 = .106$) and Miss Marple (slope = 0.078, $p = .0036$, $R^2 = .065$) followed Table 12. These results indicate a systematic erosion of character consistency as the dialogue progressed, with simulated utterances becoming increasingly complex and structurally distant from their literary origins as the investigation intensified.

Character	z Variant	Slope	R^2	Std. Err
Holmes	z_{ref}	0.105**	0.189	0.019
Holmes	z_{sim}	0.240**	0.189	0.044
Marple	z_{ref}	0.078	0.065	0.026
Marple	z_{sim}	0.140	0.065	0.047
Poirot	z_{ref}	0.093**	0.106	0.024
Poirot	z_{sim}	0.180**	0.106	0.046

Table 12: Pooled linear regression of z-scores (z_{sim}) over dialogue turns

Although the magnitude of the effect varied between detectives, the directionality of the drift was uniform across the simulation. The high slope and explanatory power for Holmes suggest that his unique syntactic profile—originally defined by a specific analytical structure—diverged most sharply from its literary baseline under the pressure of multi-agent interaction. In contrast, while Marple’s character consistency was also subject to statistically reliable drift, her syntactic deviation remained comparatively more moderate. The pooled nature of these regressions ensures that these trends reflect stable character-level patterns across different cases and experimental runs rather than isolated fluctuations.

Supplementary per-run analyses further confirmed that positive slopes predominated across individual simulations, reinforcing the robustness of this temporal effect. Taken together, these findings provide strong evidence of cumulative syntactic divergence over dialogue turns. As the collaboration advances, the agents’ structural voices increasingly migrate away from their canonical norms, further challenging the maintenance of character consistency in long-term reasoning tasks.

5.5.3 Comparison Of Reference- And Simulation-based Slopes

Character	z_{ref} Slope	z_{sim} Slope	Difference	Consistent Direction?
Holmes	0.105	0.240	0.135	Yes
Marple	0.078	0.140	0.062	Yes
Poirot	0.093	0.180	0.087	Yes

Table 13: Comparison of Reference- and Simulation-Based Temporal Slopes

Table 13 compares the temporal regression slopes obtained using the reference-based (z_{ref}) and simulation-based (z_{sim}) standardisations in order to assess the robustness of the observed syntactic drift patterns.

The results show that all three detectives exhibit positive temporal slopes under both normalisation schemes (Consistent Direction: Yes). This indicates

that the increase in syntactic complexity reflects a genuine linear evolution during dialogue generation rather than an artefact introduced by reference-based normalisation.

Among the three characters, Holmes shows the strongest syntactic drift, with a z_{sim} slope of 0.240. By contrast, Marple displays the most stable character consistency, with the smallest z_{sim} slope (0.140) among the three detectives.

5.5.4 Syntactic Results Summary

When considered collectively, the syntactic results reveal two parallel challenges to character consistency: a substantial global elevation in structural complexity relative to canonical baselines Table 11, and a progressive temporal drift as the dialogue advances 12. The uniformity of these effects suggests that role-conditioned multi-agent interaction does not merely replicate each detective’s established structural habits. Instead, the collaborative environment appears to impose an "investigative overhead", where the pressure of mystery-solving mandates increasingly complex syntactic constructions. This shift likely reflects the cognitive demands of cumulative reasoning, elaborative hypothesis building, and escalating argumentative detail required by the task.

Crucially, the consistent direction of drift across all three detectives which are characterised by positive slopes in Table 13, contrasts with the more heterogeneous patterns observed in lexical measures. This implies that while agents may retain certain unique vocabulary traits, their underlying syntactic structures are particularly sensitive to dialogue dynamics. This results in a systematic convergence towards a complex, task-oriented register that ultimately erodes the structural dimension of character consistency over time.

5.6 Discourse Metrics

This section evaluates character consistency at the discourse level, focusing on the affective dimension of the agents’ dialogue. While previous analyses examined task performance, classifier-based identity, and structural patterns (lexical and syntactic), the present analysis investigates whether agents maintain a coherent emotional profile throughout the interaction. In this study, discourse-level character consistency is operationalised through sentiment analysis using the VADER framework, which allows multi-dimensional measurement of affect across dialogue turns.

5.6.1 Comparison Of Affective Profiles With Literary Baselines

The initial assessment examines the overall alignment between simulated dialogue and the emotional signatures derived from canonical literary baselines. Table 14 summarises the average sentiment vectors for each detective in the simulation data.

Character	Positive (pos)	Negative (neg)	Neutral (neu)	Compound
Holmes	0.096	0.053	0.852	0.082
Marple	0.090	0.063	0.848	0.057
Poirot	0.084	0.057	0.859	0.038

Table 14: Average sentiment vectors for each character

Across all characters, simulated utterances are predominantly neutral ($M > 0.84$). This pattern is consistent with the tendency of LLM-generated text to display relatively neutral affective distributions. Among the three detectives, Poirot exhibits the highest level of neutrality (0.859) and the lowest compound sentiment (0.038), suggesting a particularly restrained and controlled discourse style.

The degree of discourse-level character consistency is further illustrated by the mean Euclidean distance between the simulation sentiment vectors and the literary baselines. As shown in Table 16, Holmes displays the largest mean distance ($M = 0.946$), indicating the strongest deviation from his canonical affective profile. In contrast, Marple shows the smallest distance ($M = 0.860$), suggesting that her simulated dialogue initially aligns more closely with the emotional characteristics observed in the literary corpus. Poirot occupies an intermediate position ($M = 0.886$), reflecting moderate alignment with his baseline affective style.

5.6.2 Temporal Trends In Sentiment Deviation

To examine whether discourse-level character consistency changes over time, linear regressions were fitted with turn index as the predictor and sentiment distance as the response variable. Table 15 presents the resulting slopes for the three cases.

Case	Holmes	Marple	Poirot
Case 1	0.010	0.069	0.045
Case 2	-0.005	0.055	0.015
Case 3	0.046	0.121	0.063

Table 15: Mean Slope of Sentiment Distance for Each Character

Holmes presents a different pattern. While his overall sentiment distance from the baseline is relatively high, the temporal slopes remain small. In Case 2, the slope is slightly negative (-0.005), suggesting a minor convergence toward the baseline style. However, in Case 3 the slope increases to 0.046, indicating that stronger interaction demands can still lead to gradual divergence.

Poirot again occupies an intermediate position. His slopes remain positive across all cases, but their magnitude is moderate compared with Marple. This

pattern suggests a steady but limited erosion of discourse-level character consistency during the dialogue.

Table 16 summarises the discourse-level indicators across all cases. Marple displays the largest mean slope ($M = 0.083$), indicating the strongest temporal erosion of character consistency. Holmes shows the smallest mean slope ($M = 0.018$), suggesting comparatively stable discourse behaviour over time despite his higher baseline deviation. Poirot again falls between the two, with a moderate mean slope ($M = 0.042$).

Character	Mean Slope	Std Slope	Mean Distance	Std Distance
Holmes	0.018	0.081	0.946	0.144
Marple	0.083	0.122	0.860	0.110
Poirot	0.042	0.107	0.886	0.114

Table 16: Character-Level Summary of Discourse Consistency Indicators

5.6.3 Visual Analysis Of Sentiment Distance

Visual inspection of the turn-level sentiment distances further clarifies the patterns observed in the regression analysis. Figure 4a, 4b, 5a presents the distribution of sentiment distances across dialogue turns and characters.

Across turns, all three detectives display a modest upward tendency. Sentiment distance from the baseline gradually increases as the conversation progresses. This pattern is consistent with the regression results and suggests a gradual decline in discourse-level character consistency rather than progressive stabilisation. Among the three characters, Poirot exhibits the steepest increase, whereas Holmes shows a comparatively flatter trajectory.

The boxplot Figure 5b highlights cross-character variation in sentiment distance. Marple displays the highest median distance from the baseline, followed by Poirot, while Holmes shows the lowest central tendency. In terms of dispersion, Marple also exhibits the widest distribution, indicating greater variability in emotional expression. Holmes shows moderate variability, whereas Poirot presents the most compact distribution.

Overall, the visual patterns reinforce the quantitative findings. Marple demonstrates the largest variability in discourse-level character consistency, Holmes maintains a comparatively restrained affective profile, and Poirot displays moderate sentiment intensity with relatively stable dispersion.

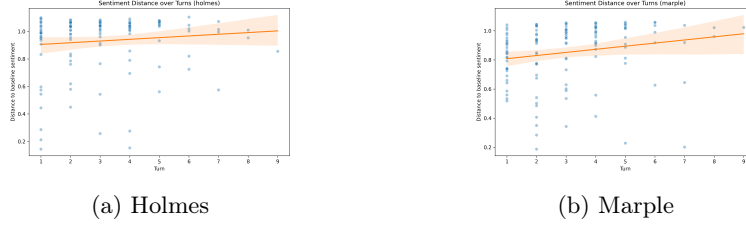


Figure 4: Plot of Sentiment Distance Over Turns For Each Character

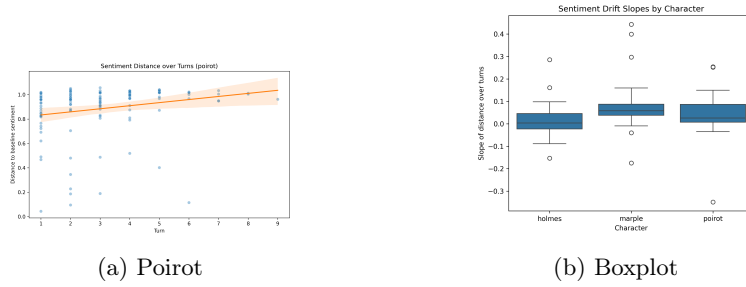


Figure 5: Plot of Sentiment Distance Over Turns For Each Character

5.6.4 Discourse Results Summary

Overall, the discourse-level results reveal a trade-off between initial alignment and temporal stability. Marple begins with the closest affective alignment to her literary baseline but exhibits the strongest temporal decline in character consistency. Holmes shows the opposite pattern: his simulated dialogue remains consistently distant from the baseline yet changes relatively little during the interaction. Poirot displays moderate baseline alignment together with a gradual but controlled increase in sentiment deviation.

Taken together, these findings suggest that discourse-level character consistency is sensitive to the collaborative dynamics. As collaborative reasoning progresses, the affective profiles of the agents tend to diverge from their literary baselines, although the magnitude and stability of this divergence vary across characters.

5.7 Cross Metric Validation

To evaluate the reliability of the proposed character-consistency measures, each turn-level metric was compared with the classifier’s probability of the correct character label. Correlation coefficients were computed to assess the association between linguistic measures and classifier confidence. In addition, expected calibration error (ECE) was calculated to examine the alignment between metric values and probabilistic predictions. The results are shown in Table 17.

Correlation and ECE were computed using all available turn-level observations. Because intra-agent similarity requires a previous utterance, some metrics (e.g. `character_distance`) are not available for the first turn of each run, resulting in slightly smaller sample sizes.

Metric	pearson_r	spearman_rho	spearman_p	ECE
lexical_similarity	0.353*	0.384**	0.226	
intra_similarity	-0.084	-0.161	0.195	
character_distance	0.018	0.102	0.130	
depth	-0.385**	-0.459**	0.301	
z_ref	-0.180	-0.244*	0.270	
z_sim	-0.321*	-0.415**	0.268	
sentiment_distance	0.183	-0.016	0.551	

Table 17: Correlation between linguistic metrics and classifier-based character consistency

5.7.1 Lexical Level

Lexical similarity shows a moderate and statistically significant positive association with classifier confidence ($r = 0.353$, $p = 0.002$; $\rho = 0.384$, $p = 0.001$). This indicates that stronger lexical alignment with the reference character profile is associated with higher classifier-based character consistency. Among the evaluated measures, lexical similarity demonstrates one of the clearest relationships with classifier predictions.

5.7.2 Syntactic Level

Syntactic depth exhibits the strongest correlation with classifier confidence ($r = -0.385$, $p = 0.001$; $\rho = -0.459$, $p < 0.001$). The negative direction indicates that increasing syntactic deviation from the reference style corresponds to lower classifier-based character consistency.

The simulation-based syntactic deviation (z_{sim}) also shows a significant negative association ($r = -0.321$, $p = 0.006$), while the reference-based measure (z_{ref}) presents a weaker relationship, reaching significance only under the Spearman correlation ($\rho = -0.244$, $p = 0.039$). These results suggest that syntactic structure provides an important signal for character identification.

5.7.3 Discourse Level

Sentiment distance does not show a statistically significant association with classifier confidence. Both linear and rank correlations remain weak, indicating limited alignment between sentiment variation and classifier-based character consistency. In the present setting, discourse-level affect therefore appears to play a relatively minor role in character recognition.

5.7.4 Comparative Faithfulness Ranking

To compare the overall reliability of the proposed measures, a composite faithfulness score was computed by combining normalised correlation strength (Pearson’s r and Spearman’s ρ) with calibration alignment ($1 - \text{ECE}$). The resulting ranking is shown in Table 18.

Syntactic depth achieves the highest faithfulness score (0.864), followed by lexical similarity (0.837) and the simulation-based syntactic deviation (z_{sim} ; 0.799). These results indicate that lexical and syntactic features provide the most reliable signals of character consistency in the current evaluation framework.

Reference-based syntactic deviation (z_{ref}) shows moderate alignment (0.541), while transition-based metrics display weaker performance. Sentiment distance ranks lowest (0.150), suggesting that discourse-level affect contributes comparatively little to classifier-based character recognition in this setting.

Metric	Faithfulness_score
depth	0.864
lexical_similarity	0.837
z_sim	0.799
z_ref	0.541
intra_similarity	0.450
character_distance	0.398
sentiment_distance	0.150

Table 18: Faithfulness Ranking of Metrics

5.8 Validation Of Character Contamination

To examine whether collaborative simulation alters character-specific linguistic representations, we conducted unsupervised clustering on turn-level embeddings produced by the fine-tuned classifier. The analysis compares the structure of simulated dialogue embeddings with those derived from the reference corpus.

K-means clustering ($k = 3$) was applied separately to reference and simulation embeddings. For the reference corpus, five bootstrap samples were drawn to match the total number of simulation utterances. Cluster quality was evaluated using the Silhouette Coefficient, cluster purity, and the Adjusted Rand Index (ARI). The resulting clustering statistics for both datasets are summarised in Table 19.

Data_type	silhouette	purity	ARI_character	ARI_case
reference	0.3437	0.8889	0.6970	NaN
reference	0.3163	0.8837	0.6846	NaN
reference	0.3368	0.8708	0.6626	NaN
reference	0.3266	0.8837	0.6885	NaN
reference	0.2973	0.8708	0.6546	NaN
simulation	0.4064	0.4005	0.0070	0.0269

Table 19: Character Contamination

Embeddings derived from the reference corpus exhibit a clear character structure. Across bootstrap samples, the Silhouette Coefficient ranges from approximately 0.30 to 0.34, while cluster purity remains consistently high (0.87–0.89). The Adjusted Rand Index with respect to character labels ranges between 0.65 and 0.70, indicating strong agreement between cluster assignments and true character identities. These results confirm that the classifier-derived embedding space preserves character-distinctive linguistic features in the reference data.

In contrast, clustering of simulation embeddings produces a substantially different pattern. Although the Silhouette Coefficient increases to approximately 0.41, cluster purity drops sharply to 0.40 and ARI with respect to character labels falls to near zero (0.007). ARI computed against case labels is also very low (0.027), indicating that the cluster structure does not correspond to the underlying case divisions.

These results suggest that collaborative simulation reorganises the embedding space: while clusters remain internally coherent, they no longer align with character identity. Instead, character-distinctive dimensions appear to weaken, producing representational convergence between characters.

Overall, the clustering analysis provides evidence of character contamination in simulation. Compared with the reference corpus, simulated dialogue substantially reduces the separability of character-specific linguistic features in the embedding space.

6 Discussion

6.1 Overview Of Findings: Addressing Research Questions

Based on the experiment results, now research questions and hypotheses could be evaluated. This section provides a summary of the findings and directly addresses the research questions and hypotheses of this study. The results show that character consistency is not a single, uniform state. Instead, it depends on which linguistic level is being measured

- **RQ1:** To what extent do LLMs maintain character consistency during collaborative reasoning tasks?

Hypothesis: LLMs will partially preserve assigned personas but will exhibit character inconsistency over longer multi-turn dialogues.

The results partially support the first hypothesis. LLM agents maintained character consistency at lexical and syntactic levels. However, this consistency was not perfect. As the collaborative dialogues progressed, agents showed systematic inconsistency. Their sentence structures became more complex and less like the original literary sources. Therefore, character consistency is partially preserved but tends to erode during long interactions.

- **RQ2:** How can multi-level linguistic metrics capture character consistency?

Hypothesis: Combining lexical, syntactic, and discourse features will better reflect persona fidelity than any single level alone.

The results support the second hypothesis. A single metric cannot fully capture the nuances of character consistency and multiple linguistic levels provides a better view. Lexical and syntactic features acted as strong character anchors. These structural patterns remained detectable even under the pressure of collaboration. In contrast, embedding-based clustering showed representational convergence. Combining these levels provides a more complete view of how agents behave in multi-agent settings

- **RQ3:** Which metrics are most reliable capture it?

Hypothesis: Metrics that correlate strongly with a dedicated persona classifier’s confidence scores (e.g. classification probability that an utterance matches the persona) will be more predictive of true consistency

The results support the third hypothesis. Syntactic depth metrics showed the strongest correlation with classifier confidence scores. Specifically, raw depth deviation was the most reliable indicator of character consistency. This confirms that structural patterns are better predictors of character fidelity than lexical or discourse features.

6.2 A Multi-level Interpretation Of Character Consistency

The findings suggest that character consistency in LLM agents is not uniform. Instead, it is a layered construct where different linguistic levels contribute unevenly to identity recognition, and some features functioning as stronger signals than others.

At lexical and syntactic level.

6.3 Limitation And Future Work

7 Conclusion

References

- Marwa Abdulhai, Ryan Cheng, Donovan Clay, Tim Althoff, Sergey Levine, and Natasha Jaques. Consistently simulating human personas with multi-turn reinforcement learning. *arXiv preprint arXiv:2511.00222*, 2025.
- Gati V Aher, Rosa I Arriaga, and Adam Tauman Kalai. Using large language models to simulate multiple humans and replicate human subject studies. In *International conference on machine learning*, pages 337–371. PMLR, 2023.
- Razan Baltaji, Babak Hemmatian, and Lav R Varshney. Persona inconstancy in multi-agent llm collaboration: Conformity, confabulation, and impersonation. *arXiv preprint arXiv:2405.03862*, 2024.
- Pranav Bhandari, Nicolas Fay, Michael J Wise, Amitava Datta, Stephanie Meek, Usman Naseem, and Mehwish Nasim. Can llm agents maintain a persona in discourse? In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 29201–29217, 2025.
- Mary Bucholtz and Kira Hall. Identity and interaction: A sociocultural linguistic approach. *Discourse studies*, 7(4-5):585–614, 2005.
- John F. Burrows. *Computation into Criticism: A Study of Jane Austen’s Novels and an Experiment in Method*. Clarendon Press, Oxford, 1987. ISBN 9780198128564.
- Hyunjoo Chae, Yongho Song, Kai Ong, Taeyoon Kwon, Minjin Kim, Youngjae Yu, Dongha Lee, Dongyeop Kang, and Jinyoung Yeo. Dialogue chain-of-thought distillation for commonsense-aware conversational agents. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5606–5632, 2023.
- Guangyao Chen, Siwei Dong, Yu Shu, Ge Zhang, Sesay Jaward, Karlsson Börje, Jie Fu, and Yemin Shi. Autoagents: The automatic agents generation framework. *arXiv preprint arXiv:2309.17288*, 2023a.
- Hongzhan Chen, Hehong Chen, Ming Yan, Wenshen Xu, Gao Xing, Weizhou Shen, Xiaojun Quan, Chenliang Li, Ji Zhang, and Fei Huang. Socialbench: Sociality evaluation of role-playing conversational agents. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2108–2126, 2024a.
- Jiangjie Chen, Xintao Wang, Rui Xu, Siyu Yuan, Yikai Zhang, Wei Shi, Jian Xie, Shuang Li, Ruihan Yang, Tinghui Zhu, et al. From persona to personalization: A survey on role-playing language agents. *arXiv preprint arXiv:2404.18231*, 2024b.

- Nuo Chen, Yan Wang, Haiyun Jiang, Deng Cai, Yuhan Li, Ziyang Chen, Longyue Wang, and Jia Li. Large language models meet harry potter: A dataset for aligning dialogue agents with characters. In *Findings of the association for computational linguistics: EMNLP 2023*, pages 8506–8520, 2023b.
- Nuo Chen, Hongguang Li, Jianhui Chang, Juhua Huang, Baoyuan Wang, and Jia Li. Compress to impress: Unleashing the potential of compressive memory in real-world long-term conversations. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 755–773, 2025.
- Junhyuk Choi, Yeseon Hong, Minju Kim, and Bugeun Kim. Examining identity drift in conversations of llm agents. *arXiv preprint arXiv:2412.00804*, 2024.
- Jonathan Culpeper. *Language and Characterisation: People in Plays and Other Texts*. Textual Explorations. Routledge, London, 2001. ISBN 978-0582357532.
- Edirlei Soares de Lima, Marco A Casanova, Bruno Feijó, and Antonio L Furtado. Characterizing the investigative methods of fictional detectives with large language models. *arXiv preprint arXiv:2505.07601*, 2025.
- Ivar Frisch and Mario Giulianelli. Llm agents in interaction: Measuring personality consistency and linguistic alignment in interacting populations of large language models. In *Proceedings of the 1st Workshop on Personalization of Generative AI Systems (PERSONALIZE 2024)*, pages 102–111, 2024.
- Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V Chawla, Olaf Wiest, and Xiangliang Zhang. Large language model based multi-agents: A survey of progress and challenges. *arXiv preprint arXiv:2402.01680*, 2024.
- Xudong Guo, Kaixuan Huang, Jiale Liu, Wenhui Fan, Natalia Vélez, Qingyun Wu, Huazheng Wang, Thomas L Griffiths, and Mengdi Wang. Embodied llm agents learn to cooperate in organized teams. *IEEE Transactions on Computational Social Systems*, 2026.
- Ke Ji, Yixin Lian, Linxu Li, Jingsheng Gao, Weiyuan Li, and Bin Dai. Enhancing persona consistency for llms’ role-playing using persona-aware contrastive learning. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 26221–26238, 2025.
- Hang Jiang, Xiajie Zhang, Xubo Cao, Cynthia Breazeal, Deb Roy, and Jad Kabbara. Personallm: Investigating the ability of large language models to express personality traits. In *Findings of the association for computational linguistics: NAACL 2024*, pages 3605–3627, 2024.
- Tian Li, Shengyuan Hu, Ahmad Beirami, and Virginia Smith. Ditto: Fair and robust federated learning through personalization. In *International conference on machine learning*, pages 6357–6368. PMLR, 2021.

- Zijun Liu, Yanzhe Zhang, Peng Li, Yang Liu, and Diyi Yang. Dynamic llm-agent network: An llm-agent collaboration framework with agent team optimization. *arXiv preprint arXiv:2310.02170*, 2023.
- Keming Lu, Bowen Yu, Chang Zhou, and Jingren Zhou. Large language models are superpositions of all characters: Attaining arbitrary role-play via self-alignment. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7828–7840, 2024.
- Suhong Moon, Marwa Abdulhai, Minwoo Kang, Joseph Suh, Widyadewi Soedarmadji, Eran Kohen Behar, and David M Chan. Virtual personas for language models via an anthology of backstories. In *Proceedings of the 2024 conference on empirical methods in natural language processing*, pages 19864–19897, 2024.
- Keyu Pan and Yawen Zeng. Do llms possess a personality? making the mbti test an amazing evaluation for large language models. *arXiv preprint arXiv:2307.16180*, 2023.
- Angelina Parfenova, Alexander Denzler, and Juergen Pfeffer. Emergent convergence in multi-agent llm annotation. In *Proceedings of the 8th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 206–225, 2025.
- Joon Sung Park, Lindsay Popowski, Carrie Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Social simulacra: Creating populated prototypes for social computing systems. In *Proceedings of the 35th annual ACM symposium on user interface software and technology*, pages 1–18, 2022.
- Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*, pages 1–22, 2023.
- Karol Przystalski, Jan K Argasiński, Iwona Grabska-Gradzińska, and Jeremi Ochab. Stylometry recognizes human and llm-generated texts in short samples. *Expert Systems with Applications*, page 129001, 2025.
- Vinay Samuel, Henry Peng Zou, Yue Zhou, Shreyas Chaudhari, Ashwin Kalyan, Tanmay Rajpurohit, Ameet Deshpande, Karthik Narasimhan, and Vishvak Murahari. Personagym: Evaluating persona agents and llms. *arXiv preprint arXiv:2407.18416*, 8(9), 2024.
- Yunfan Shao, Linyang Li, Junqi Dai, and Xipeng Qiu. Character-llm: A trainable agent for role-playing. *arXiv preprint arXiv:2310.10158*, 2023.
- Ryan Shea and Zhou Yu. Building persona consistent dialogue agents with offline reinforcement learning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1778–1795, 2023.

- Aleksandra Sorokovikova, Sharwin Rezagholi, Natalia Fedorova, and Ivan P Yamshchikov. Llms simulate big5 personality traits: Further evidence. In *Proceedings of the 1st Workshop on Personalization of Generative AI Systems (PERSONALIZE 2024)*, pages 83–87, 2024.
- Yu-Min Tseng, Yu-Chao Huang, Teng-Yun Hsiao, Wei-Lin Chen, Chao-Wei Huang, Yu Meng, and Yun-Nung Chen. Two tales of persona in llms: A survey of role-playing and personalization. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 16612–16631, 2024.
- Quan Tu, Shilong Fan, Zihang Tian, Tianhao Shen, Shuo Shang, Xin Gao, and Rui Yan. Charactereval: A chinese benchmark for role-playing conversational agent evaluation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11836–11850, 2024.
- Maryka Van Zyl and Yolande Botha. Stylometry and characterisation in the big bang theory. *Literator: Journal of Literary Criticism, Comparative Linguistics and Literary Studies*, 37(2):1–11, 2016.
- Noah Wang, Zy Peng, Haoran Que, Jiaheng Liu, Wangchunshu Zhou, Yuhan Wu, Hongcheng Guo, Ruitong Gan, Zehao Ni, Jian Yang, et al. Rolellm: Benchmarking, eliciting, and enhancing role-playing abilities of large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 14743–14777, 2024a.
- Xi Wang, Hongliang Dai, Shen Gao, and Piji Li. Characteristic ai agents via large language models. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 3016–3027, 2024b.
- Xintao Wang, Yunze Xiao, Jen-tse Huang, Siyu Yuan, Rui Xu, Haoran Guo, Quan Tu, Yaying Fei, Ziang Leng, Wei Wang, et al. Incharacter: Evaluating personality fidelity in role-playing agents through psychological interviews. In *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)*, pages 1840–1873, 2024c.
- Dekun Wu, Haochen Shi, Zhiyuan Sun, and Bang Liu. Deciphering digital detectives: Understanding llm behaviors and capabilities in multi-agent mystery games. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 8225–8291, 2024.
- Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, et al. The rise and potential of large language model based agents: A survey. *Science China Information Sciences*, 68(2):121101, 2025.
- Rui Xu, Xintao Wang, Jiangjie Chen, Siyu Yuan, Xinfeng Yuan, Jiaqing Liang, Zulong Chen, Xiaoqing Dong, and Yanghua Xiao. Character is destiny: Can

large language models simulate personadriven decisions in role-playing. *arXiv preprint arXiv:2404.12138*, 2024a.

Xinrun Xu, Yuxin Wang, Chaoyi Xu, Ziluo Ding, Jiechuan Jiang, Zhiming Ding, and Börje F Karlsson. A survey on game playing agents and large models: Methods, applications, and challenges. *arXiv preprint arXiv:2403.10249*, 2024b.

Bohao Yang, Dong Liu, Chenghao Xiao, Kun Zhao, Chen Tang, Chao Li, Lin Yuan, Guang Yang, and Chenghua Lin. Crafting customisable characters with llms: A persona-driven role-playing agent framework. *arXiv preprint arXiv:2406.17962*, 2024.

Saizheng Zhang, Emily Dinan, Jack Urbanek, Arthur Szlam, Douwe Kiela, and Jason Weston. Personalizing dialogue agents: I have a dog, do you have pets too? In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2204–2213, 2018.

Jinfeng Zhou, Zhuang Chen, Dazhen Wan, Bosi Wen, Yi Song, Jifan Yu, Yongkang Huang, Libiao Peng, Jiaming Yang, Xiyao Xiao, et al. Characterglm: Customizing chinese conversational ai characters with large language models. *arXiv preprint arXiv:2311.16832*, 2023.

Qinglin Zhu, Runcong Zhao, Bin Liang, Jinhua Du, Lin Gui, and Yulan He. Player*: Enhancing llm-based multi-agent communication and interaction in murder mystery games. *arXiv preprint arXiv:2404.17662*, 2024.

A Character Prompt

B Rule Prompt

C Case Information